Thermal Properties of Matter

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Thermal Properties of Matter

Temperature:

Temperature of a body is defined as a measure of hotness or coldness of the body. A hot pan is said to have a high temperature, and a cold pan to have a lower temperature.
An object at a higher temperature is said to be hotter than the one at a lower temperature.
Kelvin (K) is SI unit of temperature and degree Celsius (°C) is a commonly used unit.

Heat:

Transfer of energy due to temperature difference between two bodies is called heat.
Joule (J) is SI unit of heat.
Note that heat and temperature are two different quantities.

Measurement of Temperature

In order to measure the temperature of a body, we first need to identify a thermometric property of the system. A thermometric property is any physical property of the system which changes when the temperature of the body is changed.
Most commonly used thermometric property is length of a mercury column in thermometers. When the temperature of the bulb of a thermometer changes, the length of mercury column in the tube changes thereby indicating the change in temperature.

Celsius and Fahrenheit temperature scales

1. **Celsius Scale:** Andre Celsius designed the Celsius scale in 1742. He defined ice-point temperature as 0°C and the steam point temperature as 100°C. The gap between 0°C and 100°C marks was divided in 100 equal intervals.
2. **Fahrenheit Scale:** Fahrenheit designed this scale in 1717. He defined the ice-point temperature as 32°F and the steam point temperature as 212°F.

   He divided the space between 32°F and 212°F in 180 equal intervals.

   From the above definitions, we can draw out a relation between Celsius scale temperature and Fahrenheit scale temperature as:

   $$\frac{t_C - 0}{100} = \frac{t_F - 32}{180}$$

   i.e., $$t_C = \frac{5}{9}(t_F - 32)$$

   or $$t_F = \frac{9}{5}t_C + 32$$

   The normal temperature of the human body measured on the Celsius scale is 37°C which is 98.6°F.
3. **Kelvin Scale:** Kelvin scale defines 0 K at -273.15° C. 0 K is known as absolute zero temperature. Temperature below absolute zero is not possible. The intervals on a Celsius scale and a Kelvin scale are same, i.e., a temperature difference in Celsius and Kelvin scale are equal.

The Celsius and Kelvin scales differ only in the choice of zero temperature. To convert from degrees Celsius (tC) to kelvin (T) or vice-versa, we add or subtract 273.15, i.e.,

$T = t_C + 273.15$ or $t_C = T - 273.15$

where T is the symbol referring to absolute temperature.

For most numerical problems, we can round off the temperature of absolute zero to −273°C and in that case

$T = t_C + 273$ or $t_C = T - 273$

A relation for conversion from degrees Fahrenheit (°F) to Kelvin (K) or vice-versa can be given as,

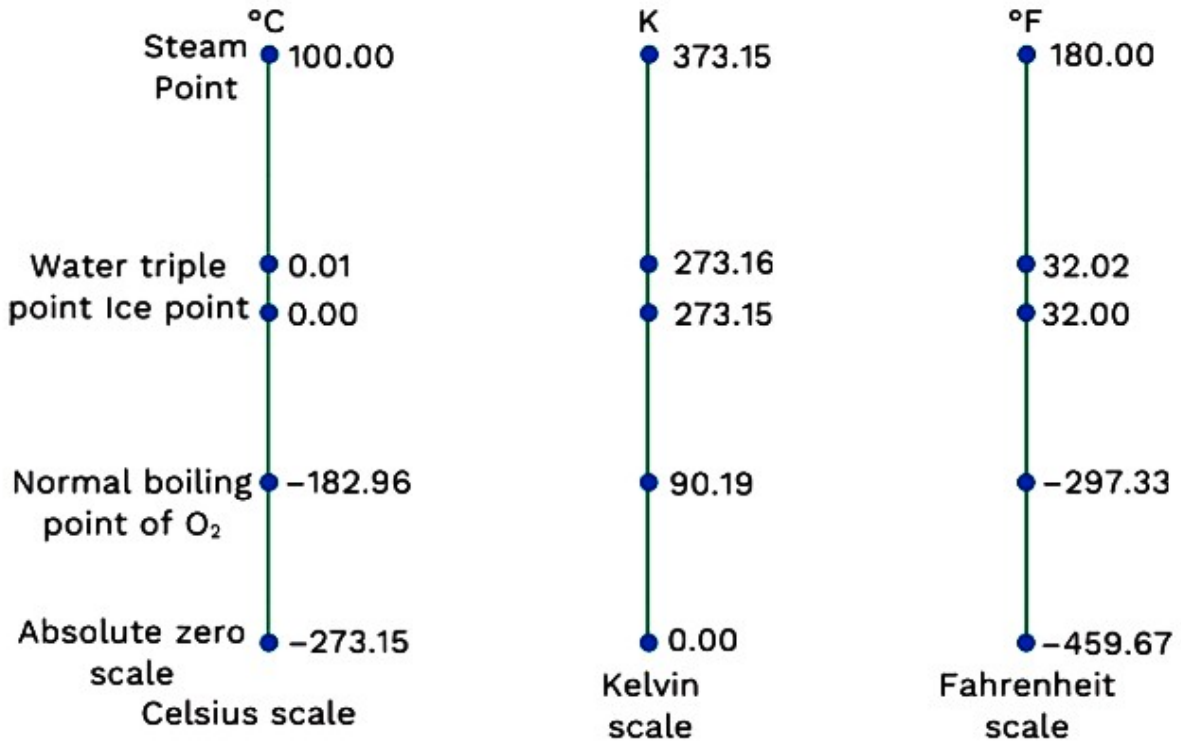
$$\frac{T - 273.15}{100} = \frac{t_F - 32}{180}$$

or $$T = \frac{5}{9}(t_F - 32) + 273.15$$

or $$T = \frac{5}{9}t_F - 17.78 + 273.15 = \frac{5}{9}t_F + 255.37$$

and $$t_F = \frac{9}{5}(T - 255.37) = \frac{9}{5}T - \frac{9}{5} \times 255.37$$

$$= \frac{9}{5}T - 459.67$$



Q. **Express the temperatures of 24.57 K and 216.55K on the Celsius and Fahrenheit scales.**

Sol. Let the corresponding temperatures on the Celsius and Fahrenheit scales be t_C & t'_C and t_F & t'_F, respectively.

Now, $t_C = T - 273.15$

$= 24.57 - 273.15 = -248.58°C$

and $t'_C = T' - 273.15 = 216.55 - 273.15 = -56.60°C$

Also, $t_F = \frac{9}{5}t_C + 32 = \frac{9}{5}(-248.58) + 32 = -447.44 + 32 = -415.44°F$

and $t'_F = \frac{9}{5}t'_C + 32 = \frac{9}{5}(-56.60) + 32$

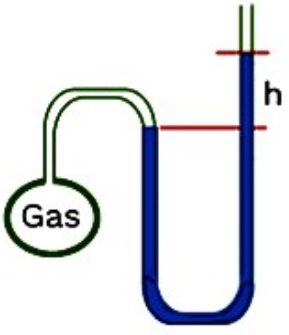
$= -101.88 + 32$

$= -69.88°F$

Two types of thermometers are mostly used, the working principles of which are discussed below:

Constant Volume Gas Thermometer

As per ideal gas law, if volume of a gas is kept constant, then the absolute pressure of the gas varies linearly with temperature. Pressure is the thermometric property here.
We can measure the pressure by the length of a mercury column as shown in the figure.

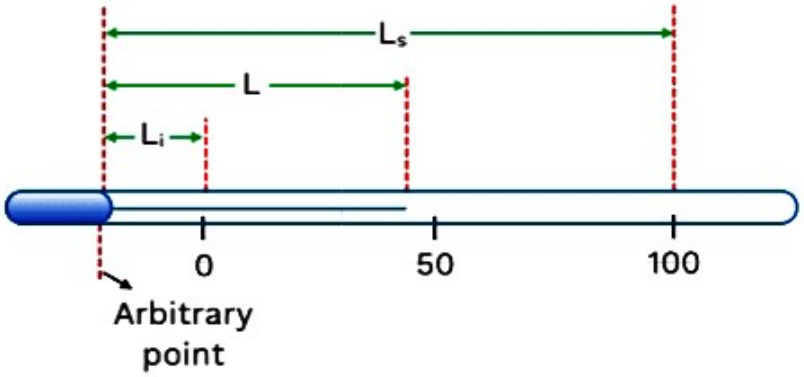


Pressure of the gas (p) is measured at ice point of water having a value pi and at steam point of water having a value ps. The ice point is taken at 0°C and steam point is taken at 100°C.

For any pressure (p), temperature will be, $T_p = (p - p_i) \times \frac{100}{p_s - p_i}$

Mercury Thermometer

The length of mercury column is the thermometric property for the mercury thermometers. Length (L) of the mercury column at ice point (L_i) and steam point (L_s) of water are measured. The ice point is taken at 0°C, and steam point is taken at 100°C.



We can find the temperature at length L as,

$$T_L = (L - L_i) \times \frac{100}{L_s - L_i}$$

Q. **A Celsius scale and Fahrenheit scale thermometer are kept in a beaker. At what temperature of the liquid in the beaker, the readings of both the thermometers are same?.**

Sol. Lets say the reading is same at temperature T, then

$$T = \frac{5}{9}(T - 32)$$

$4T = -160°$

$T = -40°$

Q. **Express a temperature of 50°F in degrees Celsius and in kelvin.**

Sol. Given that, F = 50°

So, $C = \frac{5}{9}(F - 32) = \frac{5}{9}(50 - 32)$

$= \frac{5}{9} \times 18 = 10°C$

and $K = C + 273$

$= 10 + 273$

$= 283\ K$

Q. **The temperature of a copper ball is increased from 20°C to 80°C. What is the change in its temperature expressed on the Fahrenheit scale and on the Kelvin scale?**

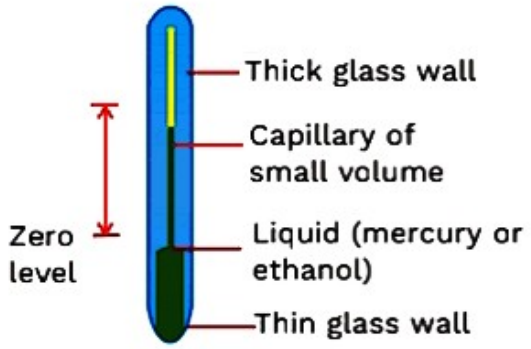
Sol. Given that, $\Delta C = 80 - 20 = 60°C$

$\Delta F = \frac{9}{5}\Delta C = \frac{9}{5} \times 60 = 9 \times 12 = 108°F$

$\Delta K = \Delta C = 60K$

Thermal Expansion

An increase in temperature causes an expansion in most of the materials.
With rise in temperature, the liquid expands in a tube of thermometer. Bimetallic strips bend due to increase in temperature. Special joints and supports are needed in bridges to allow for thermal expansion. These are few examples of thermal expansion.





Linear Expansion

When on increase in temperature, there is only a change in one dimension, then it is termed as linear expansion.

Consider that a rod of some material has a length L_0 at initial temperature T_0. If the temperature changes by ΔT, the length changes by ΔL. It can be increase or decrease in temperature causing increase or decrease in length.

It has been found experimentally that if ΔT is not very high, then

$$\Delta L \propto \Delta T, \text{ and } \Delta L \propto L_0$$

Putting a proportionality constant α (whose value is different for different materials), we get an equation.

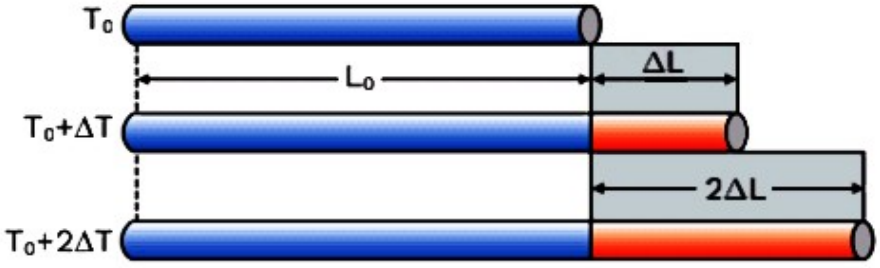
$$\Delta L = \alpha L_0 \cdot \Delta T$$

If a rod has a length L_0 at temperature T_0, then its length L at a temperature $T = T_0 + \Delta T$ is

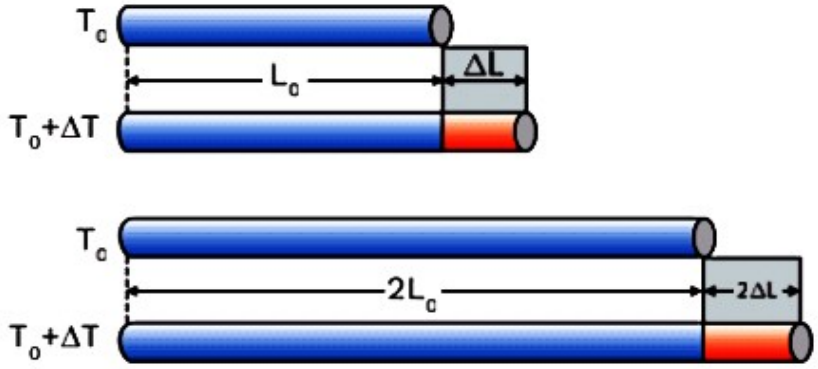
$$L = L_0 + \Delta L = L_0 + \alpha L_0 \Delta T$$

$$L = L_0(1 + \alpha \Delta T)$$

The constant α is called "coefficient of linear expansion". The unit of α is K^{-1} or $(C°)^{-1}$.



(a) For moderate temperature changes, ΔL is directly proportional to ΔT



(b) ΔL is also directly proportional to L_0

Below table gives the values of coefficient of linear expansion for some materials.

Table : Coefficients of Linear Expansion

Material	$\alpha[K^{-1}$ or $(C°)^{-1}$
Aluminium	24×10^{-5}
Brass	2.0×10^{-5}
Copper	1.7×10^{-5}
Glass	$0.4 - 0.9 \times 10^{-5}$
Invar (nickel-iron alloy)	0.9×10^{-5}
Quartz (fused)	0.04×10^{-5}
Steel	1.2×10^{-5}

Areal Expansion

If the expansion causes changes in two dimensions of the body, then it is termed as areal or superficial expansion.

We can express area as, $A = L^2$

So, $dA = 2L\,dL$

Then, we can write, $\frac{dA}{A} = \frac{2L\,dL}{L^2} = \left(2\frac{dL}{L}\right)$

Also, $dL = L_0 \alpha\, dT$

So, we have, $dA = 2(L\,\alpha\,dT)$

Replacing dA and dT with ΔA and ΔT, we get,

$\Delta A = 2\ \Delta L$

Now, if $\Delta A = \beta A\ \Delta T$, then, $\beta = 2\alpha$, where β is coefficient of areal expansion.

Volume Expansion

In volume expansion, the increase in temperature causes the change in volume.

From experimental data,

$$\Delta V = \gamma . V_0 \Delta T$$

where, γ = Coefficient of volume expansion.

Below table gives the values of γ for some materials.

Table: Thermal Coefficients of Volume Expansion

Solids	γ[K–1 or (C°)$^{-1}$]	Liquids	γ[K^{-1} or (C°)$^{-1}$]
Aluminium	7.2×10^{-5}	Ethanol	75×10^{-5}
Brass	6.0×10^{-5}	Carbon disulphide	115×10^{-5}
Copper	5.1×10^{-5}	Glycerine	49×10^{-5}
Glass	$1.2 - 2.7 \times 10^{-5}$	Mercury	18×10^{-5}
Invar	0.27×10^{-5}		
Quartz (fused)	0.12×10^{-5}		
Steel	3.6×10^{-5}		

To get a relationship between α and γ, consider a cube of material with side length L and volume $V = L^3$. Then

$$dV = \frac{dV}{dL}.dL = 3L^2 dL$$

Now we replace L and V by the initial values L_0 and V_0

$$dL = \alpha L_0 . dT$$

Since $V_0 = L_0^3$, this means that dV can also be expressed as

$$dV = 3L_0^2 \alpha L_0 dT = 3\alpha V_0 dT$$

Also, $dV = \gamma V_0 dT$, if dT is very small

$\Rightarrow \gamma = 3\alpha$

Q. **A measuring tape made of steel is exactly 50.0 m long at a temperature of 20°C. What is the length of this scale on a hot day at temperature 35°C? (Given that $\alpha_{steel} = 1.2 \times 10^{-5} K^{-1}$)**

Sol. Change in temperature, $\Delta T = T - T_0 = 15C°$,

Now, by the formula of linear expansion,

$$\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5} K^{-1})(50\,m)(15\,K) = 9.0 \times 10^{-3} m = 9.0\ mm$$

$$L = L_0 + \Delta L = 50.000\,m + 0.009\,m = 50.009\,m$$

Thus, the length of the steel tape at 35°C is 50.009 m.

Q. **If the coefficient of volume expansion of a liquid is $5.0 \times 10^{-5}\ K^{-1}$, then find the fractional change in the density of the liquid if the temperature changes by 50°C.**

Sol. We know that, density, $\rho = \frac{m}{V}$

If volume becomes V' after expansion, then

$$\rho' = \frac{m}{V'}$$

Fractional change in density, $\frac{\rho' - \rho}{\rho} = \left(\frac{\rho'}{\rho} - 1\right) = \left(\frac{V}{V'} - 1\right) = \left(\frac{V_0}{V_0 + V_0 \gamma \Delta T} - 1\right)$

$$\frac{\Delta\rho}{\rho} = -\left(\frac{\gamma \Delta T}{1 + \gamma \Delta T}\right) = -\left(\frac{(5.0 \times 10^{-5})(50)}{1 + (5.0 \times 10^{-5})(50)}\right)$$

$$= -(2.49 \times 10^{-3})$$

Q. **A beaker of glass having a volume of 200 cm³ is filled upto the brim with mercury at temperature 20°C. How much mercury will overflow when the temperature of the system becomes 100°C?**
Take $\alpha_{glass} = 0.40 \times 10^{-5}\ K^{-1}$ and $\gamma_{Hg} = 18 \times 10^{-5}\ K^{-1}$.

Sol. Coefficient of volume expansion for the glass, $\gamma_{glass} = 3\,\alpha_{glass} = 1.2 \times 10^{-5}\ K^{-1}$

Increase in volume of the glass flask can be calculated as,

$$\Delta V_{glass} = \gamma_{glass} V_0 \Delta T$$

$$= (1.2 \times 10^{-5}\ K^{-1})\ (200\ cm^3)\ (100°C - 20°C)$$

$$= 0.192\ cm^3$$

Increase in volume of the mercury will be,

$$\Delta V_{Hg} = \gamma_{Hg} V_0 . \Delta T$$

$$= (18 \times 10^{-5}\ K^{-1})\ (200\ cm^3)\ (100°C - 20°C) = 2.88\ cm^3$$

Volume of mercury that overflows,

$$V_{overflown} = \left(\Delta V_{Hg} - \Delta V_{glass}\right) = (2.88 - 0.192) = 2.688\,cm^3$$

Q. **Two rods having lengths, ℓ_1 and ℓ_2 are having coefficients of linear expansion α_1 and α_2 respectively. What will be the relation between above values, if the difference in the lengths of the two rods is independent of the variation of the temperature?**

Sol. Difference in length is $\ell = \ell_1 - \ell_2$ (assuming that $\ell_1 > \ell_2$)

As ℓ does not depend on temperature,

$$\frac{\Delta \ell}{\Delta T} = 0 \qquad \Rightarrow \frac{\Delta \ell_1}{\Delta T} - \frac{\Delta \ell_2}{\Delta T} = 0$$

Now

$$\Delta \ell = \ell \alpha\, \Delta T \qquad \Rightarrow \frac{\Delta \ell}{\Delta T} = \ell \alpha$$

So,

$$\ell_1 \alpha_1 - \ell_2 \alpha_2 = 0 \qquad \text{or} \qquad \ell_1 \alpha_1 = \ell_2 \alpha_2\text{, Which is the required relation.}$$

Q. **A change in length of metal rod A of 25 cm length is 0.050 cm when its temperature is increased from 0°C to 100°C. Another rod B of a different metal of length 40 cm expands by 0.040 cm for the same temperature rise. A third rod C having a length of 40 cm is made by joining 2 pieces of rods A and B placed end to end. This rod C expands by 0.03 cm on heating from 0°C to 50°C. What is the length of each portion of the composite rod C.**

Sol. For Rod A, $\alpha_A = \frac{\Delta\ell}{\ell\Delta T} = \frac{0.050}{25\times 100} = 2.0\times 10^{-5}C^{-1}$.

For Rod B, $\alpha_B = \frac{\Delta\ell}{\ell\Delta T} = \frac{0.040}{40\times 100} = 1\times 10^{-5}{}^{o}C^{-1}$.

Let the composite rod C has a length ℓ_1 of rod A and a length ℓ_2 of rod B.

Then, $\ell_1 + \ell_2 = 40$ ...(i)

Total expansion is

$\Delta\ell_1 + \Delta\ell_2 = 0.03$

$\Delta\ell_1 = \ell_1\alpha_A\Delta T = \ell_1 \times 2\times 10^{-5}\times 50$

$\Delta\ell_2 = \ell_2\alpha_B\Delta T = \ell_2 \times 1\times 10^{-5}\times 50$

Put these values above we get,

$(\ell_1 \times 2\times 10^{-5}\times 50) + (\ell_2 \times 1\times 10^{-5}\times 50) = 0.03$

$2\ell_1 + \ell_2 = 60$ ...(ii)

On solving (1) and (2), we get,

$\ell_1 = \ell_2 = 20\,cm$

Q. **What is the time lost by a simple pendulum per time period made up of a wire having coefficient of linear expansion α if the temperature changes by $\Delta\theta$ and the time period of the pendulum is T?**

Sol. Time period of simple pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$

or $T \propto \sqrt{l}$ ...(i)

On increasing the temperature, the length of the wire also increases.

So, $T' \propto \sqrt{l'}$ ...(ii)

From equations (i) and (ii)

$$\frac{T'}{T} = \sqrt{\frac{l'}{l}} = \sqrt{\frac{l(1+\alpha\,\Delta\theta)}{l}} = \sqrt{1+\alpha\,\Delta\theta}$$

$$\frac{T'}{T} = (1+\alpha\,\Delta\theta)^{\frac{1}{2}} \qquad [\because (1+x)^n \approx 1+nx \; x << 1]$$

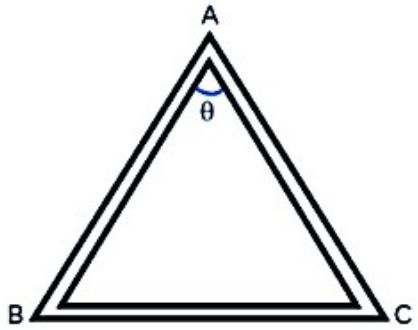
$$\frac{T'}{T} = \left[1+\frac{1}{2}\alpha\,\Delta\theta\right]$$

$$\frac{T'-T}{T} = \frac{1}{2}\alpha\,\Delta\theta$$

$$T'-T = \frac{1}{2}\alpha\,\Delta\theta\,T$$

Time lost in one time period, $T'-T = \frac{1}{2}\alpha\,\Delta\theta\,T$

Q. **The given figure shows three rods attached in the form of an equilateral triangle. All rods have same length and area of cross section. The rods AB and AC have coefficient of linear expansion 5×10^{-6}/°C, while the rod BC has coefficient of linear expansion 15×10^{-6}/°C. Find the change in angle θ when the temperature of all the rods in increased by 100°C.**



Sol. Since the three rods are identical in length so, the triangle is an equilateral triangle.
On heating the rods, rods AB and AC expand equally but the rod BC will expand more, and it will become an isosceles triangle.
Let ℓ_1, ℓ_2 and ℓ_3 be the lengths of the sides AB, BC, CA respectively, then by cosine formula $2\ell_1\ell_3\cos\theta = \ell_1^2 + \ell_3^2 - \ell_2^2$

Differentiating above equation w.r.t. temperature T,

$$2\left[\ell_1\ell_3(-\sin\theta)\frac{d\theta}{dT}+\ell_3\cos\theta.\frac{d\ell_1}{dT}+\ell_1\cos\theta\frac{d\ell_3}{dT}\right]=2\left[\ell_1\frac{d\ell_1}{dT}+\ell_3\frac{d\ell_3}{dT}-\ell_2\frac{d\ell_2}{dT}\right]$$

But when triangle in not heated, $\ell_1=\ell_2=\ell_3=\ell$ and, $\theta=60°$

$$\therefore 2\left[-\frac{\sqrt{3}\,\ell^2}{2}\frac{d\theta}{dT}+\frac{\ell}{2}\frac{d\ell_1}{dT}+\frac{\ell}{2}\frac{d\ell_3}{dT}\right]=2\ell\left[\frac{d\ell_1}{dT}+\frac{d\ell_3}{dt}-\frac{d\ell_2}{dT}\right]$$

$$-\left(\frac{\sqrt{3}\ell}{2}\right)\frac{d\theta}{dT}+\frac{1}{2}\frac{d\ell_1}{dT}+\frac{1}{2}\frac{d\ell_3}{dT}=\frac{d\ell_1}{dT}+\frac{d\ell_3}{dT}-\frac{d\ell_2}{dT}$$

Now, $\frac{d\ell_1}{dT}=\frac{d\ell_3}{dT}=\frac{d\ell}{dT}=\ell\alpha_{AB}$

and $\frac{d\ell_2}{dT}=\ell\alpha_{BC}$

$$\therefore -\frac{\sqrt{3}}{2}.\frac{d\theta}{dT}+\alpha_{AB}=2\alpha_{AB}-\alpha_{BC}$$

or $\frac{d\theta}{dT}=\frac{2}{\sqrt{3}}[\alpha_{BC}-\alpha_{AB}]$

$$\therefore d\theta=\frac{2}{\sqrt{3}}[\alpha_{BC}-\alpha_{AB}]dT$$

$$d\theta=\frac{2}{\sqrt{3}}[15\times10^{-6}-5\times10^{-6}]\times100$$

$$=\frac{2}{\sqrt{3}}\times10^{-3}\text{ radian}$$

Thermal expansion of a liquid in a container

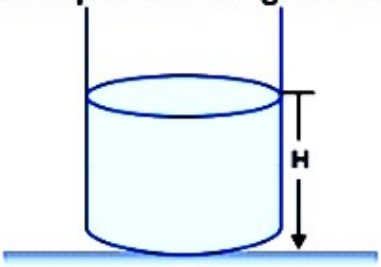
If a liquid is heated, the container in which it is kept also expands. In such cases, we need to find apparent coefficient of volume expansion. The apparent coefficient of volume expansion is given as,

$$\gamma_{apparent}=\gamma_L-\gamma_C$$

where, γ_L = coefficient of volume expansion (actual) of the liquid.

γ_C = coefficient of volume expansion of the container.

Q. **A liquid having coefficient of volume expansion γ_0 is filled in a glass vessel having coefficient of linear expansion α. When the given setup is heated to raise the temperature of the liquid and the glass container by ΔT, then**



(i) Find the relation between γ_0 and α for the fraction of volume of container occupied by the liquid to not change with temperature?

(ii) Find the relation between γ_0 and α for the level of the liquid to not change with respect to ground?

Sol. **(i)** If V_c is the volume of the container and V_L is the volume of the liquid, then as per given data,

$\frac{V_L}{V_C}$ = constant (i.e., the fraction of volume is independent of temperature)

$\Rightarrow \frac{V_L'}{V_C'} = \frac{V_L}{V_C}$ or $\frac{V_L'}{V_L} = \frac{V_C'}{V_C}$ (where V_L' and V_C' are the volumes after heating)

$\Rightarrow \frac{V_L' - V_L}{V_L} = \frac{V_C' - V_C}{V_C}$

$\Rightarrow \frac{\Delta V_L}{V_L} = \frac{\Delta V_C}{V_C}$

$\Rightarrow \gamma_0 \Delta T = (3\alpha)\Delta T$

$\boxed{\gamma_0 = (3\alpha)}$

(ii) For no change in the level of liquid wrt ground, the height H should not change.

If area of the base of the vessel is A, then

Initial volume of liquid = $V_L = A \times H$

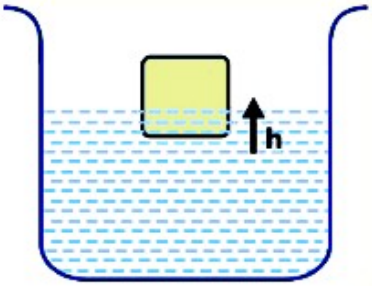
Final volume of liquid = $V_L' = A' \times H$

$\Rightarrow \frac{V_L'}{V_L} = \frac{A'}{A} \Rightarrow \frac{\Delta V_L'}{V_L} = \frac{\Delta A'}{A}$

$\Rightarrow \gamma_0 \Delta T = \beta \Delta T$

$\gamma_0 = 2\alpha$ (as $\beta = 2\alpha$)

Q. **A cube made of copper whose each side is of length 40 cm is floating on mercury as given in the figure. The density of copper cube is 3200 kg/m³ and density of mercury is 13600 kg/m³ at 27°C. By how much distance does the block sinks further when the temperature is raised from 27°C to 100°C? (Take $\gamma_{Hg} = 1.8 \times 10^{-4}$/°C and $\alpha_{Cu} = 3.0 \times 10^{-6}$/°C)**



Sol. If "m" is mass of cube, "a" is the edge length of cube and "h" is the immersed length of floating cube inside the mercury at 27°C, then

By floatation principle,

$mg = V_{in}\sigma g$ [where Vin is volume of cube inside liquid and σ = density of Hg) From the given figure, $V_{in} = h\ L^2$]

So, $m = hL^2\sigma$

As the temperature increases, the side length of cube, density of Hg and length of floating side will also change.

If side of cube is a' , density of the mercury is σ' and the length of the floating side of cube inside the liquid is h' at 100°C, then,

$m = V'_{in}\sigma'$

$= h'(L')^2\sigma'$ .. (ii)

$\therefore hL^2\sigma = h'(L')^2\sigma'$

$$\therefore \frac{h'}{h} = \left[\frac{L}{L'}\right]^2 \times \frac{\sigma}{\sigma'}$$

Now, $L' = L(1+\alpha_s\Delta T)$ and $\sigma' = \dfrac{\sigma}{(1+\gamma_L\Delta T)}$

$$\therefore \frac{h'}{h} = \frac{(1+\gamma_L\Delta T)}{(1+\alpha_s\Delta T)^2} = \frac{1+\gamma_L\Delta T}{1+2\alpha_s\Delta T}$$

$$\cong 1+(\gamma_L - 2\alpha_s)\Delta T$$

$\therefore \Delta h = h' - h = h(\gamma_L - 2\alpha_s)\Delta T$ ..... (iii)

Now we can write,

$m = hL^2\sigma$

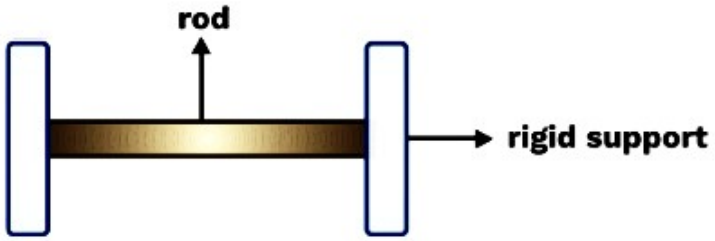
$\therefore \rho L^3 = hL^2\sigma$, where ρ is density of copper

$$\text{or } h = \frac{\rho}{\sigma}L = \frac{3200}{13600} \times 40 = 9.41\text{cm}$$

$$\therefore \Delta H = 9.41\left[(1.8\times10^{-4}) - (2\times3\times10^{-6})\right](100-27) = 0.12\,\text{cm}$$

Thermal Stress

When the expansion or contraction in a rod due to the change in temperature are prevented by external methods (like fixing both ends of the rod), then there are stresses developed in the rod. If expansion is prevented, then there is a compressive stress while there is a tensile stress if the contraction is prevented. These stresses are called thermal stresses.



Consider that a rod with length L_0 and area of cross-section A is held at constant length while the temperature is reduced (negative ΔT), causing a tensile stress. The fractional change in length if the rod was free to contract is,

$$\left(\frac{\Delta L}{L_0}\right)_{thermal} = \alpha \Delta T$$

Both ΔL and ΔT are negative. The tension must increase by an amount F that is just enough to produce an equal and opposite change in length $\left(\frac{\Delta L}{L_0}\right)_{tension}$.

By the definition of Young's modulus (Y).

$$\therefore Y = \frac{\frac{F}{A}}{\frac{\Delta L}{L_o}}$$

$$\text{So, } \left(\frac{\Delta L}{L_0}\right)_{tension} = \frac{F}{AY}$$

If the length is to be constant, the total fractional change in length must be zero.

$$\left(\frac{\Delta L}{L_0}\right)_{thermal} + \left(\frac{\Delta L}{L_0}\right)_{tension} = 0$$

$$\alpha\Delta T + \frac{F}{AY} = 0$$

$$\frac{F}{A} = -Y\alpha\Delta T$$

For a decrease in temperature, ΔT is negative, so F and $\frac{F}{A}$ are positive. This means that a tensile force and stress are needed to maintain the length.

If ΔT is positive, F and $\frac{F}{A}$ are negative, and the required force and stress are compressive.

Q. **A cylinder made of Aluminium (Y= 7.0 × 10^{10} N/m^2 and α = 2.4 × 10^{-5}/°C) having area of cross- section 20 cm^2, is to be used to keep some space between two steel walls. At 20°C, it just fits between the walls. When it is heated to 25.1°C, find the stress in the cylinder and the total force it exerts on each wall. (Assume that the walls are rigid and separation between them is constant).**

Sol. Change in temperature, $\Delta T = (25.1 - 20)C^\circ = 5.1\ C^\circ$

Thermal stress = $\frac{F}{A} = Y\alpha\Delta T$

$$= \left(7.0 \times 10^{10}\ \frac{N}{m^2}\right)\left(2.4 \times 10^{-5}\ C^{-1}\right)(5.1\ C)$$

Stress = -8.6×10^6 PA

Force on the walls, $F = \left(\frac{F}{A}\right).A$

$= (20 \times 10^{-4} m^2)(8.6 \times 10^6 Pa)$

$= 1.7 \times 10^4$ N

Calorimetry

Heat

Heat is a form of energy. Heat transfer takes place from one system to another due to a difference in temperature. If heat is added to a body, then the mechanical energy of atoms and molecules in the body increases due to their random motion.

Specific Heat Capacity

If an amount of heat Q, when given to a body of mass m, increases its temperature by an amount ΔT, then

$$Q = mc\Delta T$$

Where c is a constant and is called is the specific heat capacity or simply specific heat of the material of the material of the body.

Its value depends upon the nature of the material of the body and temperature range.

Specific heat of the material of material of a substance is the amount of heat required to raise the temperature of unit mass of the substance through 1C°.

Units Of Specific Heat : m in g and ΔT in C° and as such the unit of c is cal/gC°. In SI units, Q is measured in joule (J)m in kg and ΔT in K and as such the unit of c is J/kg K.

Molar specific heat or molar heat capacity.

If instead of unit mass, we consider one mole of a substance we talk of molar specific heat of the substance.

Molar specific heat is defined as the amount of heat required to raise the temperature of one mole of a substance through 1C°. It is denoted by C. Clearly, C = Mc, where M is the molar mass of the substance.

As $c = \frac{Q}{m\Delta T}$ and m = M × n [n is the number of moles of the substance of mass m]

$$c = M.\frac{Q}{m\Delta T}$$

$$= \frac{MQ}{Mn\Delta T}$$

$$= \frac{1}{n}\left(\frac{Q}{\Delta T}\right)$$

The SI unit of C is J/mol K.

Heat capacity or thermal capacity of body: if instead of unit mass, we consider the whole body, we talk of its heat capacity or thermal capacity.

Heat capacity of body is defined as the amount of heat required to raise the temperature of the body by one degree. It is denoted by C.

Clearly, heat capacity

$C' = mc$

$$= m\left(\frac{Q}{m\Delta T}\right) = \frac{Q}{\Delta T}$$

The SI unit heat capacity is J/K.

Water Equivalent: If the heat capacity of a body is expressed in term of mass of water, it is called water equivalent of the body.

Water equivalent of a body is the quantity of water whose temperature would be raised through 1C° (or 1K) by the same amount of heat as required to raise the temperature of the body through 1C° (or 1K).

It is denoted by w and is numerically equal to the product of mass (m) of the body and its specific heat (c). i.e.

$W = mC$ ..... (ii)

W is measured in g in cgs units and in kg in SI units.

Latent Heat : It is the amount of heat required to change the state of a unit mass of a substance without change in its temperature, and pressure.

$Q = mL$ ..... (iii)

Where L is the latent heat. Its SI units is J/kg.

It depends on

(i) Pressure, its value is generally given at standard atmospheric pressure.

(ii) Nature of the phase change, and

(iii) Properties of the substance.

The latent heat of fusion L_f is the heat gained when the phase change is from a solid to a liquid.

For water, at 1 atm pressure, latent heat of fusion value is 80 cal/g. The latent heat of vaporisation (L_v) corresponds to liquid to gas phase change. Its value for water at 1 atm is 540 cal/g.

Calorimetry studies the measurement of heat energy transferred. The vessel which is largely used in such a measurement is called a calorimeter.

Principle of Calorimetry: When two bodies at different temperatures are allowed to share heat, they attain a common temperature. If it is assumed that no heat is received from or given to any body outside the system and if there is no chemical action involved in the process of sharing, then

Heat gained = Heat lost

This statement is based on the law of conservation of energy and is called the principle of calorimetry.

Some Important Definitions:

Units of Heat: In conventional system, the unit of heat is called calorie (cal). It is the historical unit of heat and was originally defined as the amount of heat required to raise the temperature of 1 g of water through 1°C.

1 cal = 4.186 J ≈ 4.2 J

Q. An aluminium cup has a mass of 0.120 kg and is initially kept at 20.0°C. On pouring 0.3 kg of water initially at 70.0°C, find the final temperature after the water and the cup attain thermal equilibrium? (Take c_{Al} = 910 J/kg K and C_{water} = 4200 J/kg K)

Sol. $Q_{water} = m_{water}.C_{water}\Delta T_{water}$

$Q_{water} = (0.30 kg)\ (4200\ J/kgK)\ (T - 70)K$

The positive heat is gained by the aluminium cup is

$Q_{aluminium} = M_{aluminium}.C_{aluminium}\Delta T_{aluminium}$

$= (0.120 kg)\ (910\ J/kg.K)\ (T - 20.0°C)$

Now,

$Q_{water} + Q_{aluminium} = 0$

$(0.30\ kg)\ (4200)\ (T - 70) + (0.120)\ (910)\ (T - 20) = 0$

$T = 66.0°C$

Q. **A block of mass 0.2 kg made of metal at 150°C is dropped in a copper calorimeter (of water equivalent 0.025 kg) containing 150 cc of water at 27°C. The final temperature is 40°C. Calculate the specific heat of metal.**

Sol. Given mass of metal M = 0.20 kg = 200 g.

Total mass of calorimeter = (m + w) = (150 cc × 1 g cc + 0.025 × 1000)g.

(Density of water = 1 g/cc)

Fall in temperature of metal = 150 — 40 = 110°C.

Rise in temperature of water = 40 — 27 = 13°C.

Heat lost by metal = MC (110) = 22000 C

Heat gained by calorimeter = (m + w) × 1 × (13) = 2275 cal (specific heat capacity of water = 1 cal $g^{-1}{}^{\circ}C^{-1}$)

As heat lost = heat gained.

$\therefore 22000C = 2275$

Solving we get C = 0.1 cal/gm°C

Q. **A coffee machine of power rating 850 W makes 1.75 litres (10 cups) coffee at 80°C by using tap water at 20°C in 10 minutes. The percentage of electrical energy utilised to make the coffee is**

Sol. Energy required for heating of water

$Q_{out} = mc\Delta T = (1.75\ kg)\ (4200\ J/\ kg.K)\ (80 - 20)K = 441{,}000\ J$

The energy consumed by the device (electrical energy)

$W_{in} = Pt = (850\ W)\ (10\ min)\ (60\ s/min) = 510{,}000\ J$

Percentage of electrical energy used = Q_{out}/W_{in} = (441,000 J)/(510,000 J) × 100 = 86.5%

Q. **A cube made of Aluminium having a mass 100 g is removed from a bath of boiling water and dropped in a bath of room temperature water (2.00 litres at 20.0°C).**
Calculate the final temperature attained by water assuming heat loss to the surroundings is negligible? Take specific heat capacity of aluminium as 897J/kg°C

Sol. $+Q_{hot} = -Q_{cold}$

$+(mcÅT)_{aluminium} = -(mc.AT)_{water}$

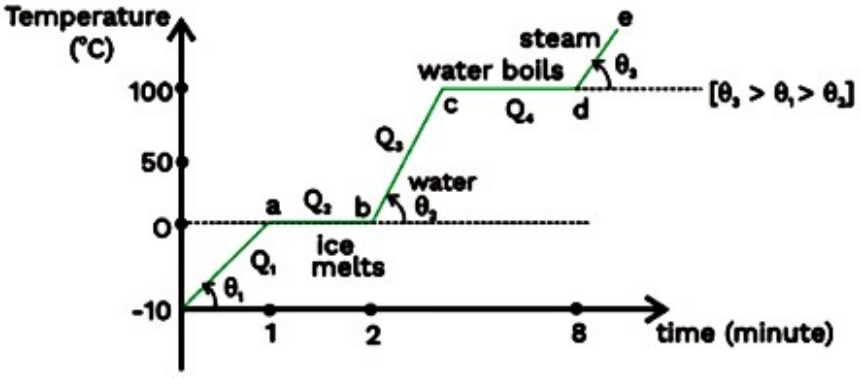
$(0.100\ kg)\ (897\ J/kg°C)\ (T_{final} - 100°C) = (2.00\ kg)\ (4182\ J/kg°C)\ (20.0°C - T_{final})$

$T_{final} = 20.8°C$

Latent heat and Phase change

All matter can exist in solid, liquid or gaseous state depending upon the values of temperature and pressure. These states or forms of matter are also called the phases of matter. It is possible to change the energy of these molecules by supplying heat which in turn can result in change of phase.

Consider a container filled with 1 kg ice at temperature (–10°C). Figure is a graph which shows the changes in the state of ice with time as more and more heat is added to it.



Up to point "a", ice remains in solid state. Heat Q_1 supplied to it raises its temperature from –10°C to 0°C.

From "a" to "b", there is no change in temperature, but the heat supplied Q_2 changes the phase from ice to water at 0°C. Q_2 is the latent heat of fusion.

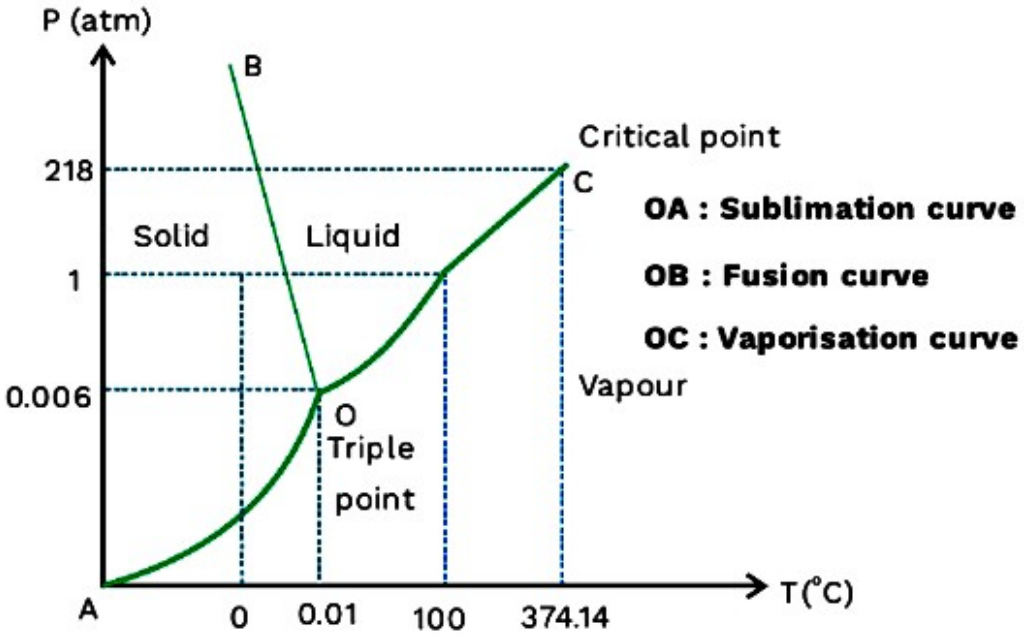
From "b" to "c", it remains in liquid state and the heat supplied Q_3 raises its temperature from 0°C to 100°C.

From "c" to "d", the temperature is constant at 100°C and the phase changes from liquid to gas (steam). Heat supplied Q_4 produces this change in phase and it is the latent heat of vapourisation.

From "d" onwards, the state is steam and the temperature increases further.

Triple point and Pressure-Temperature Diagram

(i) The boiling point of a liquid also depends upon pressure. In figure (a), curve OC shows how the boiling point of water varies with pressure (behaviour typical of other liquid as well).



A curve which shows the variation of boiling point with pressure is called the vaporisation curve. In case of water, this curve (i.e. OC) is also called steam line. The points on this curve represents the states in which liquid and vapour phases coexist.

The upper limit (C) of this curve occurs at a temperature of 374.14°C and a pressure of 218 atm and is know as the critical point. A substance cannot exist in the liquid state at a temperature above that of its critical point, regardless of how great the pressure may be. Helium has the lowest critical temperature —268°C, and is therefore, a gas at all temperatures above that.

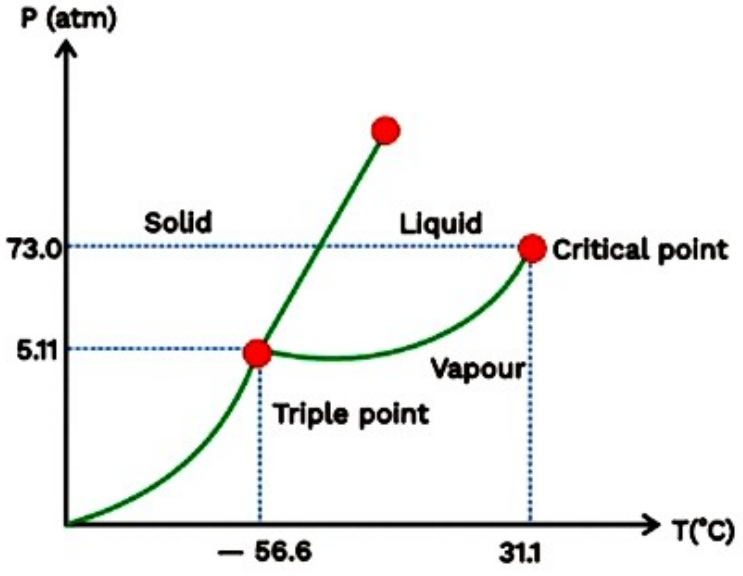
(ii) The melting point of a solid also depends upon pressure (although to a smaller extent that the boiling point). A curve which shows the variation of melting point with pressure is called the fusion curve.

In case of water, this curve is also called the ice line. In figure (a) the curve OB represents the fusion curve of water. It is to be noted that this curve leans slightly to the left. This implies that as the pressure is increased, The melting point is lowered. This behaviour of ice together with gallium and bismuth is unique as the melting point of all other substances increases with increasing pressure.

(iii) The fusion and the vaporisation curves of water intersect at a temperature of 0.01°C and at a pressure of 0.006 atm as shown in figure (a). Along the fusion curve both ice and water can simultaneously exist. Hence, under conditions corresponding to those of the intersection of the two curves, the solid, liquid and gas can all exist together. This intersection is accordingly called the triple point (O) of water.

(iv) At pressure below that of its triple point no substance can exist as a liquid. The dividing line on a pressure-temperature graph between the solid and the vapour state is called the sublimation curve since it represents the conditions required for a solid to vaporise directly or a vapour to solidify directly.

In case of water, this curve is also called hoar—frost line and is shown by OA. At atmospheric pressure, the addition of heat causes ordinary ice to melt, since the triple point of water lies at 0.006 atm which is well below. But the addition of heat causes solid CO_2 to sublime since its triple point (— 5.6°C and 5.11 atm) lies above 1 atm as shown in figure (b). Thus



Triple point of a substance is that point on the pressure temperature (P—T) diagram, whose coordinates given the particular temperature and pressure at which the solid, liquid and vapour phases of the substance coexist in equilibrium without any change in their proportions.

Q. **250 g of ice at temperature 0°C is added to 600 g of water at 18°C kept in a vessel.**

(a) Find the final temperature attained by the system?

(b) How much ice remains when thermal equilibrium is reached?

Take:

Specific heat capacity of water : 4190 J/K.kg

Specific heat capacity of ice : 2100 J/K.kg

Latent heat of fusion of ice : 3.34×10^5 J/kg

Sol. We first need to check whether complete ice will melt or not as per the given data. If heat supplied by the water in cooling down 0°C. is less than sufficient to melt ice, some ice will remain in the mixture and final temperature will be 0°C. Heat required to melt ice is

$Q_{required} = mL$

$= (250 \times 10^{-3}) \times 3.34 \times 10^5 = 8.35 \times 10^4$ J

Heat available from water is

$Q_{available} = Mc\Delta T = 600 \times 10^{-3} \times 4190 \times 1187$

$= 4.525 \times 10^4$ J

As $Q_{available} < Q_{required}$, so whole ice does not melt. Find temperature is 0°C. Mass of ice remaining, $m = \dfrac{(8.35 - 4.525) \times 10^4}{3.34 \times 10^5} = 0.114$ kg

= 114 gm

Q. **100 g of ice at –10°C is to be converted into steam at 100°C. The specific heat of ice at –10°C is 420 cal/kg and latent heat of vaporisation of water is 542 kcal/kg. The ice is first heated with a 350 watt heater for 4 minute and after this it is heated with an Infra-red lamp of λ = 9000 Å for 30 minute at an efficiency of 40%. Find the rate at which the photons are striking the ice when heated with Infra-red lamp.**

Sol. First of all, it is required to find the total amount of heat required to convert ice at –10°C into steam at 100°C. Heat absorbed by ice to change its temperature from –10°C to 0°C.

$Q_1 = mC\Delta T$

$= 0.1 \times 0.42 \times 10 = 0.42$ kcal

Heat absorbed by ice at 0°C to convert into water at 0°C

Q_2 = mL = 0.1 × 10 = 8 kcal

Heat absorbed by water to charge its temperature from 0°C to 100°C

Q_3 = mCΔT = 0.1 × 1 × 100 = 10 kcal

Heat absorbed by water at 100°C to convert into steam at 100°C

Q_4 = mL = 0.1 × 542 = 54.2 kcal

So, total heat absorbed by ice at −10°C in converting it into steam at 100°C

Q = 0.42 + 8 + 10 + 54.2

= 72.62 kcal = 72.62 × 4.2 × 10^3

= 305.0 kJ

Now, heat supplied by 350 W heater in 4 minutes

Q_H = 350 × 4 × 60 = 84 kJ

∴ Heat supplied by IR lamp = Q − Q_H

= 305.0 − 84

= 221.0 kJ (1)

Let n photons strike per second on ice,

∴ The energy of incident photons in time t is given as E = nhνt

or $E = \frac{nhc}{\lambda}t$ [∵ c = νλ]

As t = 30 min = 30 × 60 = 1800 s

$$\therefore E = n\frac{6.62\times10^{-34}\times3\times10^{8}\times1800}{9000\times10^{-10}}$$

$$= n\frac{6.62\times3\times18}{9\times10^{-7}}\times10^{-24}$$

$$= n\times39.72\times10^{-17}\,J$$

Now efficiency of lamp is 40%.

$$\therefore Q_{IR} = n\times39.72\times10^{-17}\times\frac{40}{100}$$

$$= n\times15.88\times10^{-17}\,J \quad (2)$$

∴ From (1) and (2)

$$n\times15.88\times10^{-17} = 221\times10^{3}$$

or $$n = \frac{221\times10^{3}}{15.88\times10^{-17}}$$

$$= 1.39\times10^{21} \text{ photons/second}$$

Q. **500 g of petroleum was burnt in a bomb calorimeter which contains 5 kg of water and has water equivalent 300 g. The rise in temperature was 10°C. Determine the calorific value of petroleum.**

Sol. Heat absorbed by the water

$Q_1 = mC\Delta T = 5000 \times 1 \times 10 = 50$ kcal

And heat absorbed by calorimeter

$Q_2 = WC\Delta T = 300 \times 1 \times 10 = 3$ kcal

$\therefore$ Total heat absorbed $Q = Q_1 + Q_2$

$Q = 50 + 3 = 53$ kcal

As heat is produced by combustion of 500 g of fuel,

$\therefore$ Calorific value of petroleum

$$= \frac{53 \times 10^3}{500} = 106\,\text{cal / g} = 0.106\ \text{kcal / g}$$

Heat Transfer

We know that heat transfer takes place due to a temperature difference. Now, we will see the mechanisms through which heat transfer takes place.

Heat transfer can occur through 3 mechanisms, namely conduction, convection and radiation.

Conduction occurs between two bodies kept in direct contact or within a body.

Convection occurs due to motion of mass carrying heat from one region of space to another.

Transfer of heat by electromagnetic radiations is termed as Radiation, e.g heating with sunlight which does not need any matter to be present in between bodies.

Conduction :

If you hold one end of a copper rod and place the other end in a flame, the end you are holding gets hotter and hotter, even though it is not in direct contact with flame. Heat reaches the cooler end by conduction through the material. On the atomic level, the atoms in the hotter regions have more kinetic energy, on the average than their cooler neighbours. They warm their neighbours, giving them some of their energy. The neighbours warm their neighbours, and so on through the material. The atoms themselves do not move from one region of material to another, but their energy does.

Most metals also uses another, more effective mechanism to conduct heat. Within the metal, some electrons can leave their parent atoms and wander through the crystal lattice. These 'free' electrons can rapidly carry energy from the hotter to the cooler regions of the metals, so metals are generally good conductors of heat.

Heat transfer occurs only between regions that are at different temperatures, and the direction of heat flow is always from higher to lower temperature. Figure (a) below shows a rod of conducting material with cross sectional area A and length L. The left end of the rod is kept at a temperature T_1 and the right end at a lower temperature T_2, so heat flows from left to right. The sides of the rod are covered by an insulator, so no heat transfer occurs at the sides.

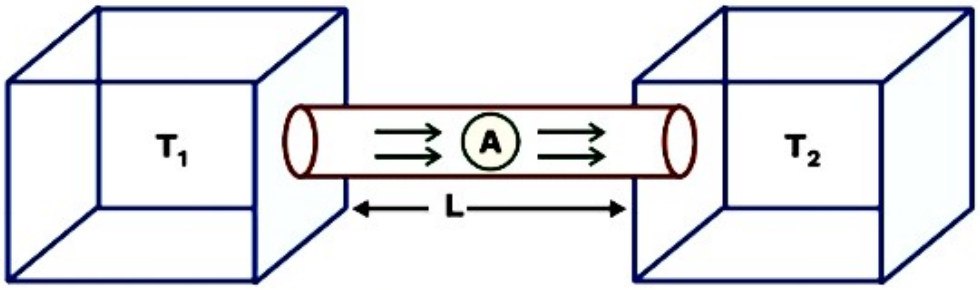


Fig. (a) Steady state heat flow due to conduction in a uniform rod

When a quantity of heat dQ is transferred through the rod in a time dt, the rate of heat flow is dQ/dt. We call this rate the heat current, denoted by H. That is, $H = \frac{dQ}{dt}$. Experiments show that the heat current is proportional to the cross-sectional area A of the rod as in figure (b) and to the temperature difference $(T_1 - T_2)$ and is inversely proportional to the rod length L as in figure (c).

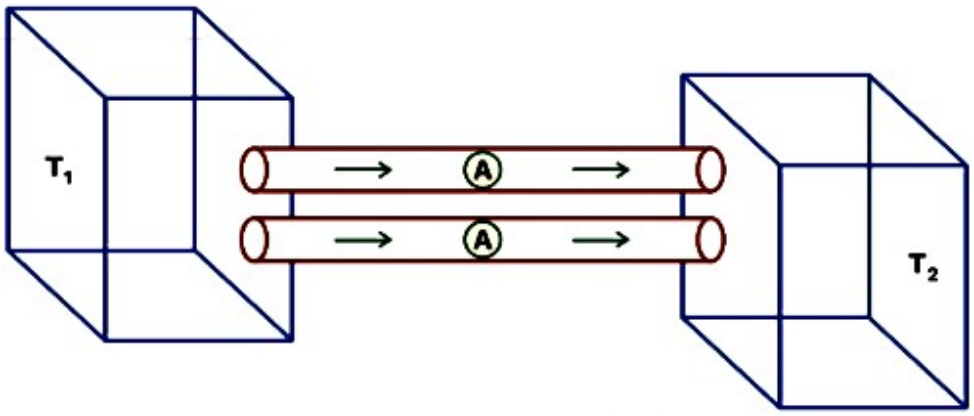


Fig. (b) Doubling the cross-section area of the conductor doubles the heat current ($H \propto A$)

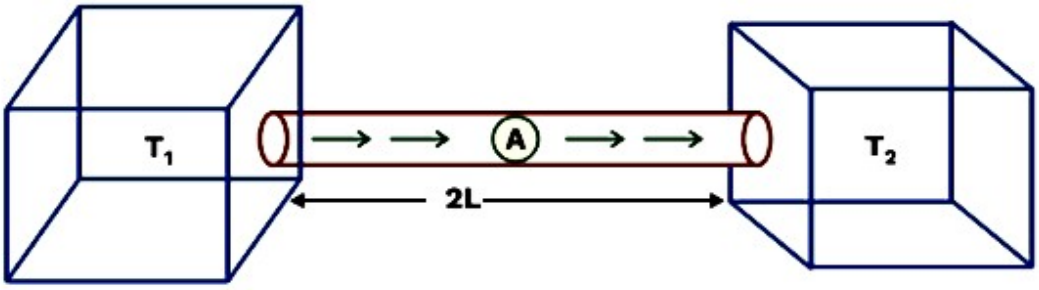


Fig. (c) Doubling the length of the conductor halves the heat current $\left(H \propto \frac{1}{L}\right)$

Introducing a proportionality constant K called the thermal conductivity of the material, we have

$$H = \frac{dQ}{dt} = \frac{KA}{L}(T_1 - T_2) \qquad (1)$$

The quantity $\frac{(T_1 - T_2)}{L}$ is the temperature difference per unit length. It is called the magnitude of the **temperature gradient**. The numerical value of K depends on the material of rod. Materials with large K are good conductors of heat, material with small K are poor conductors or insulators. Equation (1) also gives the heat current through a slab or through any homogeneous body with uniform cross section A perpendicular to the direction of flow, L is the length of the heat-flow path.

Table : Thermal Conductivities (K)

Substances	K(W/m.K)
Metals	
Aluminium	205.0
Brass	109.0
Copper	385.0
Lead	34.7
Mercury	8.3
Silver	406.0
Steel	50.2
Solids (representative values)	
Brick, insulating	0.15
Brick, red	0.6
Concrete	0.8
Cork	0.04
Felt	0.04
Fiberglass	0.04
Glass	0.8
Ice	1.6
Rock wool	0.04
Styrofoam	0.01
Wood	0.12 – 0.04
Gases	
Air	0.024
Argon	0.016
Helium	0.14
Hydrogen	0.14
Oxygen	0.023

The unit of heat current H or power is unit of energy per unit time. The SI unit of heat current is the wat (1 W = 1 J/s). We can find the unit of K by using equation (1). For K, we can show that the SI units are W/mK. The thermal conductivity of "dead" (that is non-moving air) is very small. A wool sweater keeps you worm because it traps air between the fibers. In fact many insulating materials such as Styrofoam and fiberglass are mostly dead air.

If the temperature varies in a non-uniform way along the length of the conductor rod, we introduce a coordinate x-along the length and generalise the temperature gradient to be $\left(-\frac{dT}{dx}\right)$. The corresponding generalisation of equation (1) is

$$H = \frac{dQ}{dt} = -KA\frac{dT}{dx} \qquad (2)$$

The negative sign shows that heat always flows in direction of decreasing temperature.

Fourier's Law of Heat Conduction

The heat transfer rate across the cylindrical section is given by Fourier's law as

$$Q = \frac{dH}{dt} = -kA\frac{dT}{dx}$$

Where,

Q is heat transfer rate (unit : W)

$\frac{dT}{dx}$ is temperature gradient in the direction of the flow (unit : K/m)

k is the thermal conductivity of the material (unit : W/mK)

A is the cross-sectional area of heat path

Therefore, the heat transfer rate by conduction through the object in the above figure can be expressed as :

$$Q = -kA\frac{\Delta T}{\Delta x} = kA\frac{(T_1 - T_2)}{L}$$

Where,

A – Cross-sectional area of the object

L – Wall thickness

$T_1 - T_2$ is temperature difference between two surfaces

k – Thermal conductivity of object's material

Steady State Heat Conduction

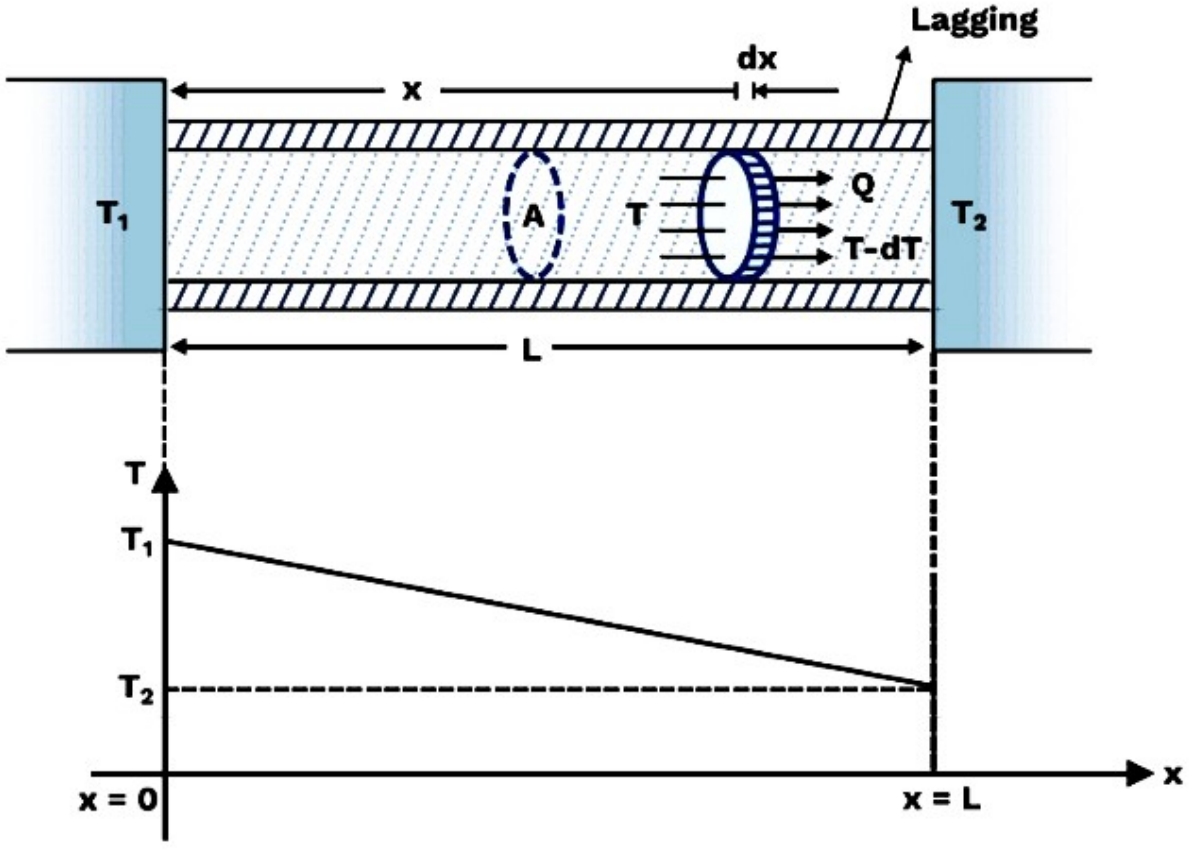
Consider an arrangement of two reservoirs having infinite heat capacity and temperatures T_1, T_2. The two reservoirs are connected by a rod having length L, area of cross-section A and uniform thermal conductivity k.

Let us suppose that initially the rod was at room temperature and it is just connected to the reservoirs. The temperature of the rod starts increasing (becomes unsteady). At this state, the rod absorbs heat. After some time, the temperature stops changing with time (become steady). At this situation, the heat in flow from the hot reservoir is equal to the heat out flow to the cold reservoir. Also, the rate of heat flow through any cross section is same, i.e., Q is same everywhere.

Let Q be the heat flow rate through a cross-section at distance x and of thickness dx. Using Fourier's law of heat conduction,

$$Q = -kA\frac{dT}{dx}$$

or $$dT = -\frac{Q}{kA}dx$$



Integrating within proper limits,

$$\int_{T_1}^{T_2} dT = -\frac{Q}{kA}\int_0^L dx$$

$$T_2 - T_1 = -\frac{Q}{kA}L$$

$$\Rightarrow\ Q = \frac{kA}{L}(T_1 - T_2)$$

Temperature distribution across the rod :

As $\quad Q = -kA\frac{dT}{dx}$

$$dT = -\frac{Q}{kA}dx$$

Integrating with limits

$$\int_{T_1}^{T} dT = -\frac{Q}{kA}\int_0^x dx$$

$$\Rightarrow\ T - T_1 = -\frac{Q}{kA}x$$

or $\quad T - T_1 = -\frac{x}{L}(T_1 - T_2)$

$$\Rightarrow\ T = T_1 - \frac{x}{L}(T_1 - T_2)$$

The variation has been plotted above.

Thermal Resistance

The heat transfer by conduction due to temperature difference has an analogy with flow of electric current through a wire when a potential difference is applied. In that case, electrical resistance is defined as

$$R = \frac{v}{i}$$

Similarly, thermal resistance is defined as

$$R = \frac{(T_1 - T_2)}{Q}$$

For a rod having length L, area of across-section A and thermal conductivity k.

$$R = \frac{T_1 - T_2}{Q}$$

$$= \frac{(T_1 - T_2)}{kA(T_1 - T_2)/\ell}$$

$$\Rightarrow R = \frac{\ell}{kA}.$$

Having calculated the thermal resistance, we can now apply the results of series combination and parallel combination of resistors. It has been explained below.

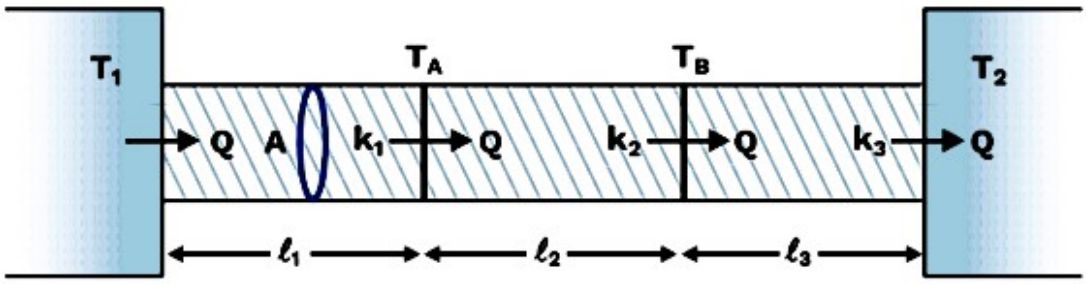
Series Combination :

$$R = \Sigma R_i$$

$$R = \frac{\ell_1}{k_1 A} + \frac{\ell_2}{k_2 A} + \frac{\ell_3}{k_3 A}$$

The heat flow rate is given by

$$Q = \frac{(T_1 - T_2)}{R}$$



The temperature of the junctions can be calculated as given below

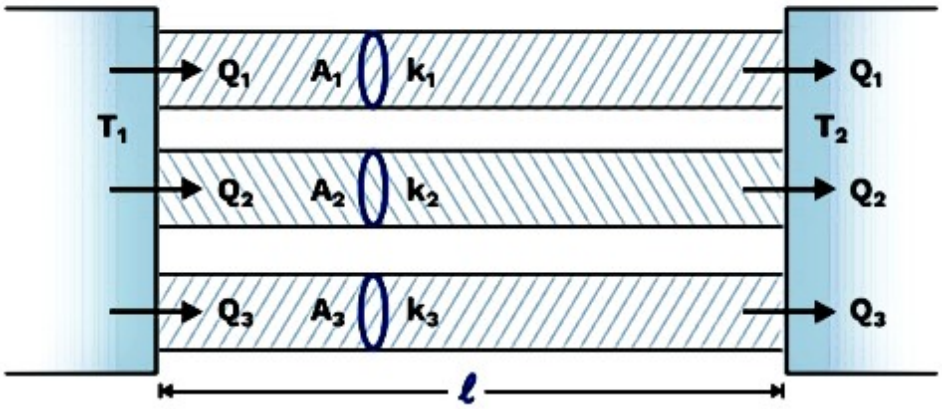
$$Q = \frac{T_1 - T_A}{R_1} = \frac{T_A - T_B}{R_2} = \frac{T_B - T_2}{R_3} = \frac{T_1 - T_2}{R}$$

$$\Rightarrow T_A = \frac{T_1(R_2 + R_3) + T_2 R_1}{R_1 + R_2 + R_3};\ T_B = \frac{T_1 R_3 + T_2(R_1 + R_2)}{R_1 + R_2 + R_3}$$

Parallel Combination :

Here $\frac{1}{R} = \Sigma \frac{1}{R_i}$

$$\Rightarrow \frac{1}{R} = \frac{k_1 A_1}{\ell} + \frac{k_2 A_2}{\ell} + \frac{k_3 A_3}{\ell}$$



Total heat flow rate between the reservoirs is

$$Q = Q_1 + Q_2 + Q_3 = \frac{T_1 - T_2}{R_1} + \frac{T_1 - T_2}{R_2} + \frac{T_1 - T_2}{R_3}$$

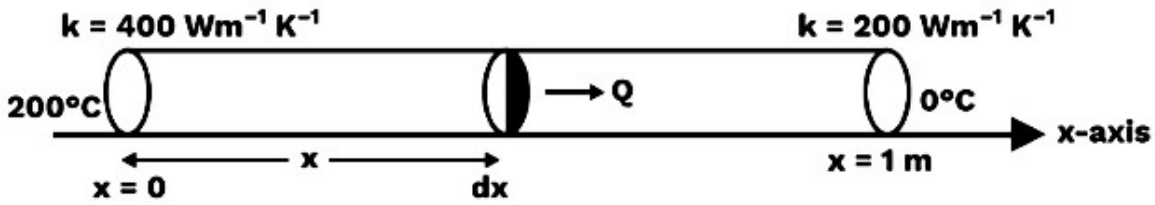
Also,

$$Q = \frac{(T_1 - T_2)}{R}$$

Q. **Due to a spatial variation in purity, the thermal conductivity of a metal bar (cross sectional area 4×10^{-4} m², length 1m) decreases linearly along its length from 400 $Wm^{-1} K^{-1}$ at one end, to 200 $Wm^{-1} K^{-1}$ at the other. Calculate the rate at which heat flows through the bar if the hot end is maintained at 200°C and the cold end at 0°C.**

Sol. As the thermal conductivity is not same throughout the thickness, you cannot apply Fourier's law for the metal bas as a whole. Let one end of the bar be at x = 0 and the bar is oriented along x-axis. As the conductivity decreases linearly from 400 $Wm^{-1} K^{-1}$ to 200 $Wm^{-1} K^{-1}$, the conductivity at the cross-section which lies at a distance x from origin is

$$k = 400 - \left(\frac{400 - 200}{1}\right) x$$



The rate of heat flow through the section of thickness dx will be

$$Q = -kA\frac{dT}{dx} = -(400 - 200x) \times 4 \times 10^{-4}\frac{dT}{dx}$$

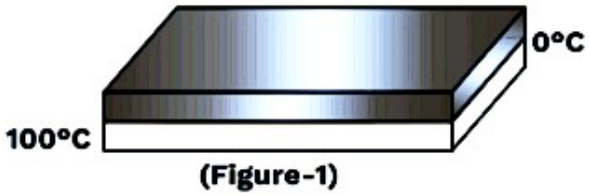
$$\Rightarrow -dT = \frac{Qdx}{4 \times 10^{-4}(400 - 200x)}$$

$$\Rightarrow -\int dT = \frac{Q}{4 \times 10^{-4}}[\ln(400 - 200x)]_0^1 \times \frac{1}{(-200)}$$

$$\Rightarrow -(0 - 200) = \frac{Q}{4 \times 10^{-4} \times 200}\ln(2)$$

or $$Q = \frac{200 \times 4 \times 10^{-4} \times 200}{\ln 2} = 23.08\,W$$

Q. **Two identical rectangular rods of metal are welded as shown in figure (1) and 20 J of heat flows through the rods in 1 min. How long would it take for 20 J heat to flow through the rods if they are welded as shown in figure (2).**



(Figure-1)



(Figure-2)

Sol. Let R be the thermal resistance of each rod.

$\therefore$ In first case $\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R}$ or $R_p = \frac{R}{2}$

So the rate of flow of heat in this situation will be

$$\frac{dQ}{dt} = \frac{\Delta T}{R_p} = \frac{100 - 0}{R/2} = \frac{20}{60}$$

or $R = \frac{200 \times 60}{20} = 600\,°C/W$

Now for case (2)

$R_s = R + R = 600 + 600 = 1200\,°C/W$

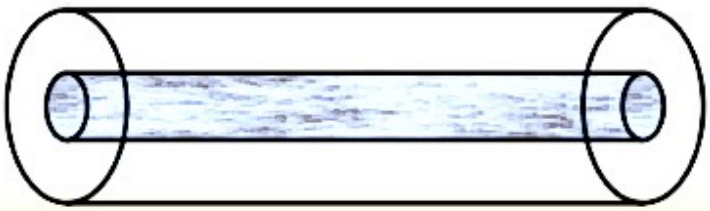
$\therefore \frac{dQ}{dt} = \frac{\Delta T}{R_s}$

$\Rightarrow \frac{20}{t} = \frac{100}{1200}$

$\Rightarrow t = 20 \times 12 = 240\,s$

or $t = 4$ min.

Q. **A cylinder of length 1 m is made of two materials. The inner core of cylinder of radius r is made of a material of thermal conductivity K_1 and the outer shell of inner radius r and outer radius 2r is made of material of thermal conductivity K_2. If we assume that there is not loss of heat across the cylindrical surface and the two ends of system are maintained at temperature difference ΔT, then what is the effective thermal conductivity of the system?**



Sol. The length of inner rod = 1 m

The area of cross-section of inner rod = πr^2

Let Q_1 be the amount of heat passing through the rod in time t, when its ends are at temperature difference ΔT.

$\therefore$ According to Fourier's law of conduction

$$\frac{dQ_1}{dt} = k_1(\pi r^2)\frac{\Delta T}{1} = k_1 \pi r^2 \Delta T \qquad (1)$$

The length of outer shell is also 1 m

The area of cross-section of outer shell $= \pi(2r)^2 - \pi r^2 = 3\pi r^2$

$\therefore$ Rate of flow of heat $\frac{dQ_2}{dt} = k_2(3\pi r^2)\cdot\frac{\Delta T}{1}$

or $\frac{dQ_2}{dt} = k_2 3\pi r^2 \Delta T$ (2)

Let k be the equivalent conductivity

$\therefore \frac{dQ}{dt} = k(\pi(2r)^2)\frac{\Delta T}{1}$

or $\frac{dQ}{dt} = k\,4\pi r^2 \Delta T$ (3)

From equation (1) and (2)

$\frac{dQ}{dt} = \frac{dQ_1}{dt} + \frac{dQ_2}{dt} = k_1\pi r^2\Delta T + k_2 3\pi r^2 \Delta T$ (4)

$\therefore$ From (3) and (4)

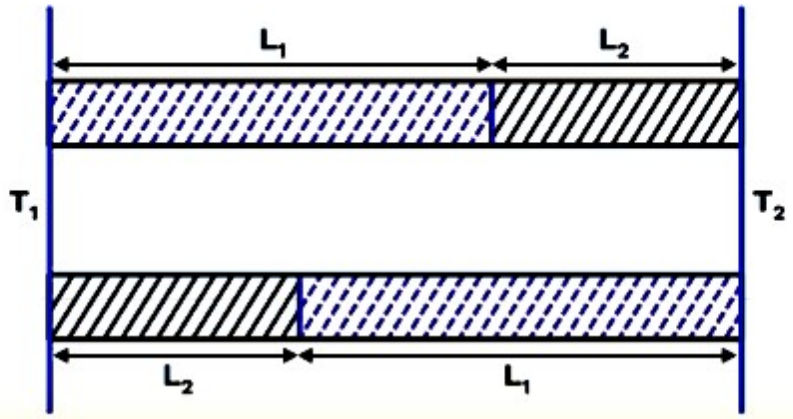
$k_1\pi r^2 + k_2 3\pi r^2 = k4\pi r^2$

or $4k = k_1 + 3k_2$

or $k = \frac{k_1 + 3k_2}{4}$

$\therefore$ Equivalent conductivity of cylinder $= \frac{3k_2 + k_1}{4}$

Q. Two rods each of cross-section area A, length L_1 and L_2 and thermal conductivities k_1 and k_2 respectively are joined to form a single rod as shown in figure. If two such rods parallel to each other have temperature T_1 and T_2 at their free ends, then find the thermal resistance of the combination in terms of k_1 and k_2.



Sol. The law of thermal conduction

$$Q = kA\frac{(T_1 - T_2)}{L}t$$

or $$\frac{Q}{t} = \frac{kA}{L}(T_1 - T_2)$$

or can be written as $H = \frac{\Delta T}{R}$ with $H = \frac{Q}{t}$ and $\Delta T = T_1 - T_2$.

Where $R = \frac{L}{kA}$ (thermal resistance)

Now, if two rods are in series, then the heat flowing per second through them will be $H_1 = \frac{T_1 - T}{R_1}$ and $H_2 = \frac{T - T_2}{R_2}$, where T is temperature at the joint of rods.

But in series $H_1 = H_2 = H$

$\therefore$ $T_1 - T = HR_1$ and $T - T_2 = HR_2$

Adding these equations we get

$$T_1 - T_2 = H(R_1 + R_2)$$

or $$H = \frac{T_1 - T_2}{R_1 + R_2} = \frac{\Delta T}{R_1 + R_2}$$

or $H = \frac{\Delta T}{R}$ with $R = R_1 + R_2$

$\therefore$ In series the total resistance is equal to the sum of individual resistances.

Now, if two rods are parallel with each of resistance $R_1 + R_2$, then

$$H_1 = \frac{\Delta T}{R_1 + R_2} \text{ and } H_2 = \frac{\Delta T}{R_1 + R_2}$$

or $$H = H_1 + H_2 = \frac{2\Delta T}{R_1 + R_2} = \frac{\Delta T}{R}$$

where $R = \frac{R_1 + R_2}{2}$

But $R_1 = \frac{L_1}{k_1 A}$ and $R_2 = \frac{L_2}{k_2 A}$

$\therefore$ The thermal resistance of combination

$$= \frac{L_1}{2k_1 A} + \frac{L_2}{2k_2 A} = \frac{1}{2A}\left[\frac{k_1 L_2 + k_2 L_1}{k_1 k_2}\right]$$

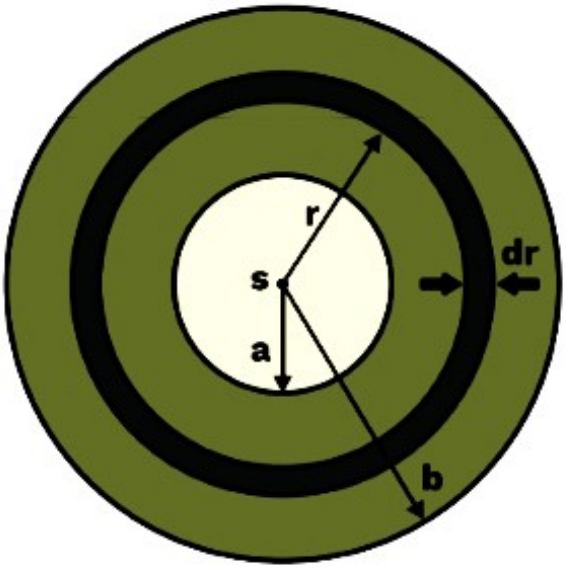
Q. **A point source of heat is placed at the centre of a spherical shell of mean radius 10 cm. The material of the shell has thermal conductivity 9.5×10^{-2} kcal/m-s°C. Calculate the power of point source of heat, when the temperature difference between the outer and inner surface of the shell is 50°C in steady state. The thickness of shell is 2 cm.**

Sol. Consider a concentric spherical shell of radius r and thickness dr as shown in figure.

The radical rate of flow of heat through this shell in steady state will be

$$Q = -\frac{dH}{dt} = -kA\frac{dT}{dr}$$

The negative sign is used as with increase in r, T decreases.



Now

$A = 4\pi r^2$

$$\therefore \quad Q = -4\pi r^2 k\frac{dT}{dr}$$

or $$\frac{dr}{r^2} = -\frac{4\pi k}{Q}dT$$

Let a and b be inner and outer radius of shell and T_1 and T_2 be temperature of inner and outer surface.

$$\therefore \quad \int_a^b \frac{dr}{r^2} = -\frac{4\pi k}{Q}\int_{T_1}^{T_2} dT$$

or $$\left[-\frac{1}{r}\right]_a^b = -\frac{4\pi k}{Q}[T_2 - T_1]$$

$$\frac{1}{a} - \frac{1}{b} = -\frac{4\pi k}{Q}(T_2 - T_1)$$

or $$Q = k\frac{4\pi ab(T_1 - T_2)}{(b-a)}$$

Here $T_1 - T_2 = 50°C$ and $k = 9.5 \times 10^{-2}$ kcal / m-s°C.

Now in steady state no heat is absorbed, rate of loss of heat by conduction is equal to that of supply i.e.

Q = power of the source

$$\therefore \quad P = k\frac{4\pi ab(T_1 - T_2)}{b-a}$$

Now, mean radius of shell is 10 cm

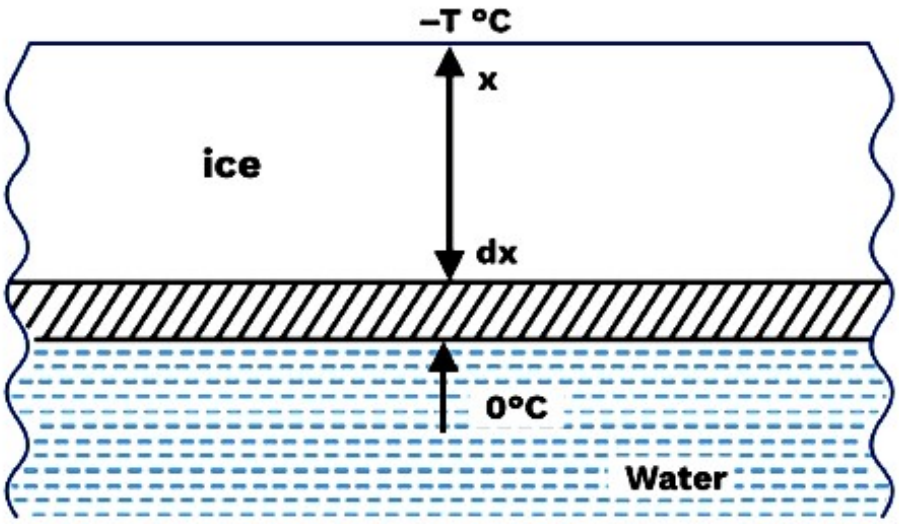
$\therefore$ a = 9 cm, b = 11 cm and b – a = 2 cm

$$\therefore \quad P = \frac{9.5 \times 10^{-2} \times 10^3 \times 4\pi \times (99) \times 50 \times 10^{-4}}{2 \times 10^{-2}}$$

$$= 5.90 \times 10^4 \text{ cal / s}$$

Growth of Ice in Pond

When temperature falls below 0°C, then pond starts freezing from the top and the thickness of ice layer grows downward towards bottom of lake. Let us consider pond has frozen upto thickness x at time t and in time dt, dx thickness grows. The heat released in freezing dx layer will be conducted out through the thickness of ice already frozen.



We know that

$Q = mL$

$$\therefore \quad \frac{dQ}{dt} = L\frac{dm}{dt} = L\rho \cdot A\frac{dx}{dt} \qquad \text{(A is area of surface)}$$

Also, $$\frac{dQ}{dt} = \frac{K_{ice}A(T)}{x}$$

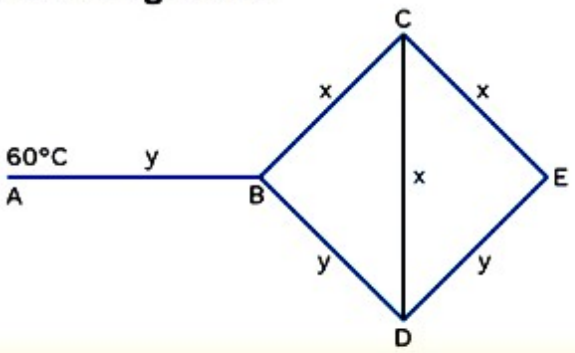
$$\frac{K_{ice}A(T)}{x} = L \cdot \rho A\frac{dx}{dt}$$

$$\frac{K_{ice}AT}{L\rho}\int_0^t dt = \int_{x=0}^{x=x} x\,dx$$

$$t = \frac{L\rho}{2K_{ice}T}x^2$$

The time taken to increase the thickness from x_1 to x_2 is $t = \frac{L\rho}{2K_{ice}T}(x_2^2 - x_1^2)$

Q. **Three rods of material x and three rods of material y are connected as shown in figure. All the rods are of identical length and cross-sectional area. If the end A is maintained at 60°C and the junction E at 10°C, calculate the temperature of the junction B, C and D. The thermal conductivity of x is 0.92 cgs units and that of y is 0.46 cgs units.**



Sol. Let T_B = Temperature of junction B
T_C = Temperature of junction C
T_D = Temperature of junction D
ℓ = Length of each rod
From junction rule, we have

$$\frac{dQ_{AB}}{dt} = \frac{dQ_{BC}}{dt} + \frac{dQ_{BD}}{dt}$$

$$\therefore \quad \frac{0.46A(60 - T_B)}{\ell} = \frac{0.92A(T_B - T_C)}{\ell} + \frac{0.46A(T_B - T_D)}{\ell}$$

or $\quad 4T_B - 2T_C - T_D - 60 \qquad (1)$

Similarly, $\frac{dQ_{BC}}{dt} = \frac{dQ_{CD}}{dt} + \frac{dQ_{CE}}{dt}$

$$\frac{0.92\,A(T_B - T_C)}{\ell} = \frac{0.92\,A(T_C - T_D)}{\ell} + \frac{0.92\,A(T_C - 10)}{\ell}$$

or $\quad T_B - 3T_C + T_D = -10 \qquad (2)$

Again, $\frac{dQ_{BD}}{dt} + \frac{dQ_{CD}}{dt} = \frac{dQ_{DE}}{dt}$

or $\quad \frac{0.46\,A(T_B - T_D)}{\ell} + \frac{0.96\,A(T_C - T_D)}{\ell} = \frac{0.46\,A(T_D - 10)}{\ell}$

or $\quad T_B + 2T_C - 4T_D = -10 \qquad (3)$

On solving, we get

$T_B = 30°C$ and $T_C = T_D = 20°C$

Thus, Temperature of the junction B = 30°C
and Temperature of C and D = 20°C

Convection

Convection is the transfer of heat by mass motion of a fluid from one region of space to another. There are two types of convection.

(i) **Forced convection :** If the fluid is circulated by an external agent like a blower of pump, the convection process is known as forced convection. The most important mechanism for heat transfer within human body is forced convection of blood, with the heart serving as the pump

(ii) **Natural convection or free convection :** If the flow is caused by difference in density due to thermal expansion, such as hot air rising, the process is called natural convection or free convection. Free convection in the atmosphere plays an important role in determining the daily weather, and convection in the oceans is an important global heat transfer mechanism.

Convection heat transfer is very complex process, and there is no simple equation to describe it. Here are a few experimental facts :

(i) The heat current due to convection is directly proportional to the surface area. This is the reason for the large surface areas of radiators and cooling fans.

(ii) The viscosity of fluids slows natural convection near a stationary surface, giving a surface film that on a vertical surface typically has about the same insulating value as 1.3 cm of plywood (R value = 0.7). Forced convection decreases the thickness of this film, increasing the rate of heat transfer. This is the reason for the "wind chill factor", you get cold faster in a cold wind than in still air with the same temperature.

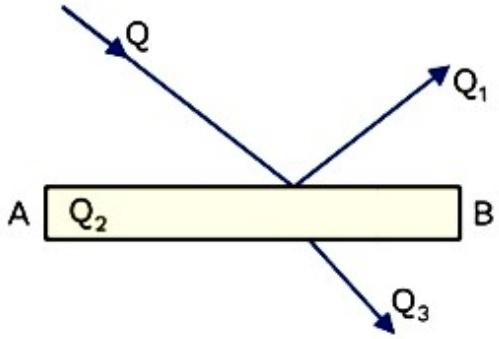
(iii) The heat current due to convection is found to be approximately proportional to the $\frac{5}{4}$ power of the temperature difference between the surface and the main body of fluid.

Radiation

Radiation is the transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet rays. Everyone has felt the warmth of the sun's radiation and intense heat from a charcoal grill or the glowing coals in a fireplace. Most of the heat from these bodies reaches you not by conduction or convection in the intervening air but radiation. This heat transfer would occur even if there were nothing but vacuum between you and the source of heat.
Everybody, even at ordinary temperatures, emits energy in the form of electromagnetic radiation. At ordinary temperature, say 20°C, nearly all the energy is carried by infrared waves with wavelengths much longer than those of visible light. As the temperature rises, the wavelength shift to shorter values. At 800°C a body emits enough visible radiation to be self-luminous and appears "red hot", although even at this temperature most of the energy is carried by infrared waves. At 3000°C, the temperature of an incandescent lamp filament, the radiation contain enough visible light that the body appears "white-hot".

Good Absorbers are Bad Reflectors

Let Q be the radiant energy incident on the surface AB as shown in Figure



$Q \rightarrow$ incident energy
$Q_1 \rightarrow$ reflected energy
$Q_2 \rightarrow$ absorbed energy
$Q_3 \rightarrow$ transmitted energy

If out of this Q incident energy, Q_1 is reflected, Q_2 is absorbed and Q_3 is transmitted, then

$\frac{Q_1}{Q} = r$ (reflected of the surface)

$\frac{Q_2}{Q} = a$ (absorptance of the surface)

$\frac{Q_3}{Q} = t$ (transmittance of the surface)

None of these quantities (r, a and f) has a unit as these are pure ratios

Further, $r + a + t = \frac{Q_1}{Q} + \frac{Q_2}{Q} + \frac{Q_3}{Q} = 1$

Thus, $r + a + t = 1$

In case, the surface does not transmit radiation, $t = 0$ and as $r + a = 1$

Obviously, if a is more, r is less or vice-versa. Thus, good absorbers are good reflectors.

Blackbody

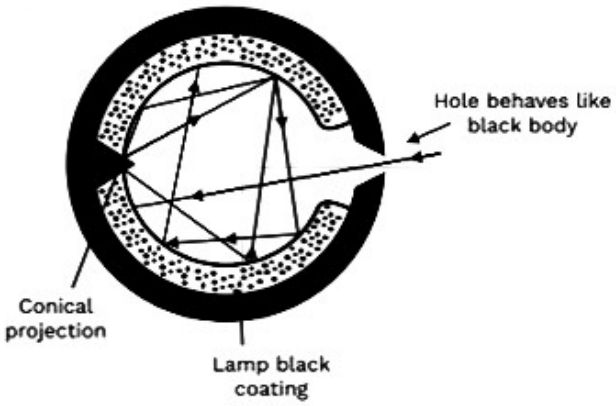
A perfect blackbody is one which absorbs radiation of all wavelengths incident on it. Its absorptance is unity as it neither reflects nor transmits any radiations.

We know that the colour of an opaque body depends upon the wavelength of radiation reflected by it. Since a blackbody reflects no wavelength, it appear black, whatever be the colour of the incident radiation. On the other hand, when such a body is heated, it emits radiation of all possible wavelength. The radiation given out by a perfect blackbody is called Blackbody radiation.

A blackbody does not mean that its colour is black, although a black-coloured body may come close to being a blackbody. Examples of black bodies, Fery's black body, the sun.

Fery's blackbody

The radiation inside an enclosure whose inner walls are maintained at a constant temperature has the same properties as the black body radiation.



Most of the energy entering through hole is absorbed (≈ 99%)

Newton's Law of Cooling

It is a common observation that bodies at temperatures higher than that of the surroundings lose heat. Newton was the first to make a scientific study in this direction.

From experimental observations, he came to a conclusion which is known as the Newton's law of cooling according to which :

The rate of loss of heat of a body is directly proportional to the excess of the temperature $(T - T_0)$ of the body with respect to the surroundings. $\frac{-dT}{dt} \propto (T - T_0)$

The rate of loss of heat also depends upon the nature and area of the radiating surface. Normally, the experiments are carried out in still air and the law is valid for a temperature difference of about 30°C only.

However, when the air around the body is in continuous motion, the law holds good even for higher temperature differences.

Verification of Newton's law of Cooling : The law is verified in an indirect way : It is assumed to be correct and the result so obtained is verified experimentally.

As per the statement of the law, $-\frac{dQ}{dt} \propto (T - T_0)$

or $-\frac{dQ}{dt} = K(T - T_0)$ (1)

Here, K is a constant of proportionality which is positive and its value depends upon the area and nature of the radiating surface.

As $dQ = mc\,dT$, $\frac{dQ}{dt} = mc\frac{dT}{dt}$ (2)

From equations (1) and (2),

$$-mc \cdot \frac{dT}{dt} = K(T - T_0)$$

or $\frac{dT}{T - T_0} = -\frac{K}{mc}dt = -kdt$ (3)

where $k = \frac{K}{mc}$ = a constant

Integrating both sides of equation (iii), we get

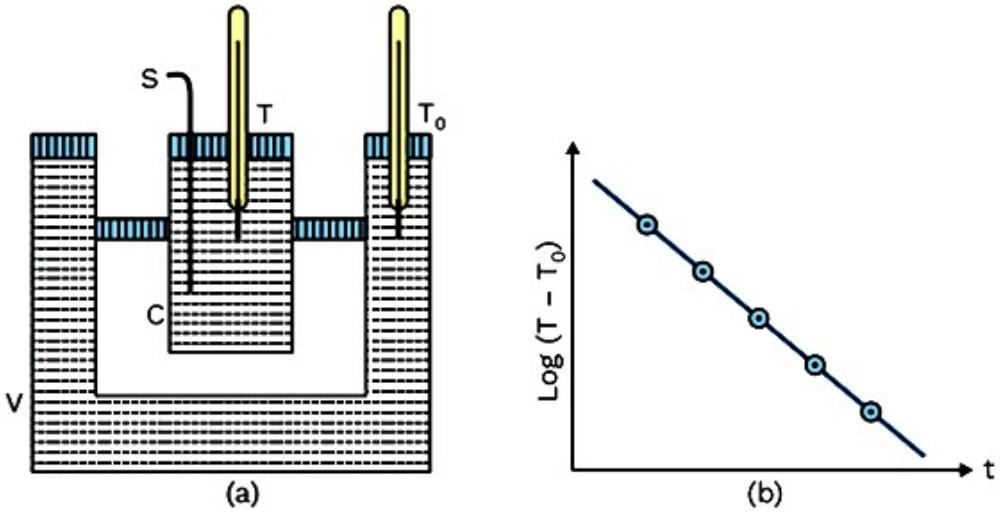
$$\int \frac{dT}{T - T_0} = -k\int dt$$

or $\ln (T - T_0) = -kt + C$ (4)

Where C is constant of integration

Equation (4) represents a straight line. Therefore, for Newton's law of cooling to be true, the graph between $\ln(T - T_0)$ and t should be a straight line.

Experiment arrangement : The apparatus used is shown in figure. It consists of a copper calorimeter (C) blackened from outside (so as to ensure the same nature of its entire radiating surface) and fitted with a cork containing a half degree thermometer (T) and a stirrer (S). Water heated to about 30°C to 40°C above the room temperature is put in the calorimeter. It is then suspended in a double walled vessel V. The space between the wall of this vessel is filled with water at room temperature. This serves as a constant temperature enclosure. Another half degree thermometer (T_0) is used to measure the temperature of the enclosure.



Water is gently stirred and its temperature (T) is noted regularly first after every half minute, then after every minute and towards the end after every two minutes. The process is repeated till the temperature of water in the calorimeter is about 5°C above the mean temperature (T_0) of the enclosure. The values of $(T - T_0)$ are calculated for various values of time t. A graph is then plotted between $\ln(T - T_0)$ and t. The graph is found to be a straight line as shown in figure (b) above, thereby verifying Newton's law of cooling.

Stefan's Law

The rate P_r at which an object radiates energy is proportional to the fourth power of the absolute temperature and is given by Stefan's law :

$$P_r = \frac{Q}{\Delta t} = \sigma A \varepsilon T^4$$

Here

1. P_r is the power radiated
2. A is the surface area of the object.
3. $\sigma = 5.67 \times 10^{-8}\ w m^{-2} K^{-4}$ is the Stefan-Boltzmann constant.
4. ε is the emissivity, which is a dimensionless number between 0 and 1, characteristic of the material.
   (i) Dark surface have emissivities close to 1
   (ii) Shiny surface have emissivities close to 0
   (iii) The emissivity of the human skin is about 0.7
   (iv) Dark surfaces are not only better emitters of radiation, they are also good absorbers.
   (v) Shiny surfaces are poor absorbers, since most of the incident radiation is refracted.
   (vi) Hence, it is better to wear light-coloured clothes in the summer, and dark coloured clothes in the winter.

Emissive Power (e)

This is defined as energy radiated per second per unit area of the surface or in other words, it is the intensity of radiation. Its S.I. unit is W/m^2 or $J\ s^{-1}\ m^{-2}$. This is also known as radiant emittance.

Emissive power of a surface depends on its nature and temperature. It is maximum for perfectly black body and minimum for a smooth shining white body.

Spectral Emissive Power (e_λ)

The emissive power corresponding to a particular wavelength is called spectral emissive power. This is also known as spectral radiant emittance. It is denoted by e_λ and is related to emissive power through the relation

$$e = \int_0^\infty e_\lambda d\lambda$$

Absorptive Power (a)

This is defined as the ratio of the radiant energy absorbed by it in a given time to the total radiant energy incident on it in the same time. Absorptive power is dimensionless and has no units.

Spectral Absorptive Power (a_λ)

This is defined as the ratio of the radiant energy of given wavelength absorbed by a given surface in given time to the total radiant energy of that wavelength incident in the same time on the same surface with a unit wavelength range. It is denoted by a_λ and is related to absorptive power through the relation

$$a = \int_0^\infty a_\lambda d\lambda$$

Emissivity (ε) : It is defined as emissive power of surface to the emissive power of perfectly back body.

$$\varepsilon = \frac{e}{E}, 0 \le \varepsilon \le 1$$

It is unitless and depend on finishing of surface

If, $\varepsilon = 1$, surface is perfectly black

$\varepsilon = 0$, surface is fully reflected

$0 < \varepsilon < 1$, surface is ordinary

Kirchhoff's Law

In general, a good emitter is also a good absorber.

The ratio of the spectral emissive power ($e\lambda$) to the spectral absorptive power ($a\lambda$) for a given wavelength λ is same for all bodies and this value is equal to the spectral emissive power of a black body at the same temperature.

Mathematically we can write,

$$\frac{e_\lambda}{a_\lambda} = e_{\lambda b}$$

Black Body

An ideal or perfect absorber (and emitte) is referred to as a black body ($\varepsilon = 1$)

Cooling of a Hot Body

When an object is in thermal equilibrium with its surroundings (environment), its temperature is constant emits and absorbs radiation at the same rate.

The rate P_a at which an object absorbs energy via thermal radiation from its environment, taken to be at a uniform temperature T_{env} is

$$P_a = \sigma \varepsilon A T_{env}^4$$

If the temperature of the object and is environment / surroundings are different, there must be a net fig radiant energy.
If an object is at temperature T and its environment is at temperature T_{env}, then the net rate of energy or gain per unit time is given by

$$P_{net} = P_a - P_r = \frac{Q}{\Delta t} = \sigma A\varepsilon(T_{env}^4 - T^4)$$

If $T_{env} < T$, then P_{net} is negative, indicating a net energy loss.
Heat can be transferred in the absence of a temperature difference, if $T_{env} = T$, there is continuous exchange of radiant energy, but there is no net change of internal energy.

Newton's Law of Cooling

Let T = Temperature of body
T_s = Temperature of surrounding
Further, we assume $T = T_s + \Delta T$, where ΔT is small. According to Stefan's law, rate of heat loss is given by

$$\frac{dQ}{dt} = A\varepsilon\sigma(T^4 - T_s^4)$$

$$= A\varepsilon\sigma T_s^4\left[\left(1+\frac{\Delta T}{T_s}\right)^4 - 1\right]$$

$$\approx 4\,A\varepsilon\sigma T_s^4\,\Delta T \quad \text{(Using binomial theorem)}$$

$$\text{Also } \frac{dQ}{dt} = -ms\frac{dT}{dt}$$

$$\therefore \quad -\frac{dT}{dt} = \frac{4\,A\varepsilon\sigma}{ms}T_s^3\Delta T$$

$$\text{or} \quad -\frac{dT}{dt} \propto \Delta T$$

$$\text{Let } \frac{dT}{dt} = -k(T - T_s)$$

$$\Rightarrow \frac{dT}{T - T_s} = -kdt$$

$$\Rightarrow \int_{T_i}^{T}\frac{dT}{T - T_s} = -\int_0^t kdt$$

$$\Rightarrow \log_e(T - T_s)\Big|_{T_i}^{T} = -k(t)_0^t$$

$$\Rightarrow \log_e(T - T_s) - \log_e(T_i - T_s) = -kt$$

$$\Rightarrow \log_e\left[\frac{T - T_S}{T_i - T_S}\right] = -kt$$

$$\Rightarrow \frac{T - T_S}{T_i - T_S} = e^{-kt}$$

$$\Rightarrow (T - T_S) = (T_i - T_S)e^{-kt}$$

$$\Rightarrow T = T_S + (T_i - T_S)e^{-kt}$$

This gives the time taken by a body to cool down from T_1 to T_2.
Hence, we can say the rate of cooling is proportional to temperature difference between the body and its surroundings provided the temperature difference is small. This is called Newton's Law of Cooling.

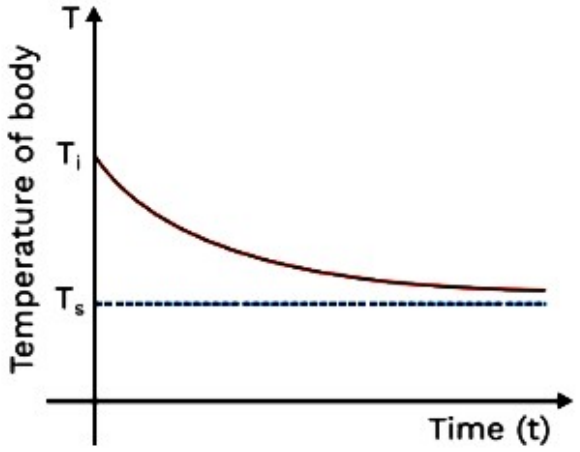


Figure shows cooling curve of a body initially at temperature T kept in a surroundings of temperature T_s. If a body cools from T_1 to T_2 in time t, then taking the approximation

$$-\frac{dT}{dt} = \frac{T_1 - T_2}{t} \text{ and } T = T_{av} = \frac{T_1 + T_2}{2}$$

The equation $\left(-\frac{dT}{dt}\right) = \alpha(T - T_S)$ becomes

$$\boxed{\frac{T_1 - T_2}{t} = \alpha\left(\frac{T_1 + T_2}{2} - T_S\right)}$$

Q. **A body cools in 10 minutes from 60°C to 40°C. What will be its temperature after next 10 minute ? The temperature of the surroundings is 10°C.**

Sol. According to Newton's law of cooling

$$\frac{T_1 - T_2}{t} = \alpha\left[\frac{T_1 + T_2}{2} - T_s\right]$$

For the given conditions, $\frac{60 - 40}{10} = \alpha\left[\frac{60 + 40}{2} - 10\right]$ (1)

Let T be the temperature after next 10 minutes. Then

$$\frac{40 - T}{10} = \alpha\left[\frac{40 + T}{2} - 10\right] \quad (2)$$

Solving equation (i) and (ii), we get

T = 28°C

Applications

1. Heat is transferred from the Sun to the Earth through (nearly) empty space by means of electromagnetic waves.
2. If you stand near an open fire, you can feel the heat on you exposed hands and face. The heat transfer is not due to convection or conduction, since heated air rises and air is a poor conductor. Visible radiation is emitted from the burning material, but most of the heating effect comes from the invisible infrared radiation emitted by the glowing ember or coals.
3. Red infrared lamps are used to keep food warm in cafeterias.
4. Although infrared radiation in invisible to the human eye, it can be detected by other means. The frequency of infrared radiation is proportional to the temperature of its source. This is the basis for infrared thermometers, which using infrared detectors, can measure temperature remotely.
5. **The Microwave Oven :** The microwave oven is a common kitchen appliance which is both time saving and energy-saving, since the oven doesn't have to be warmed up like conventional oven. The principle of operation of the microwave oven is heat transfer by radiation.
6. **The Thermo Flask :** The thermos bottle, sometimes called a Dewar flask, is designed to minimise heat transfer by conduction, convection and radiation. It is used to store either hot or cold liquids for long periods of time. The standard vessel is double-walled Pyrex glass with a silvered inner wall. The space between the walls is evacuated to minimise transfer by conduction and convection. By reflecting most of the radiant heat, the silvered surface minimizes heat transfer by radiation. Very little heat is lost through the neck of the flask because Pyrex glass is a poor conductor. A further reduction in heat loss is achieved by reducing the size of the neck.

7. Wool sweaters and down jackets keep us warm by trapping the warmer air in regions close to our bodies and hence reducing heat loss by convection and conduction. In other words, what keeps us warm is not the clothing itself but the air trapped in the clothing.

Q. **A body cools in 7 minute from 60°C to 40°C. What will be its temperature after the next 7 minute ? The temperature of surroundings is 10°C ?**

Sol. Here, temperature difference is small hence, Newton's law cooling can be used. Let T is the required temperature

$$7 = \frac{1}{K}\log_e\left[\frac{60-10}{40-10}\right]$$

$$7 = \frac{1}{K}\log_e\left[\frac{40-10}{T-10}\right]$$

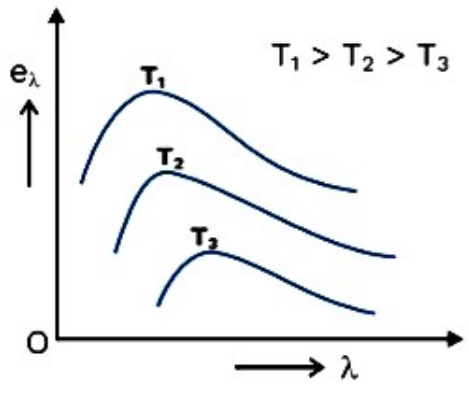
From above two equations, we get

$$\log_e\left[\frac{50}{30}\right] = \log_e\left[\frac{30}{T-10}\right]$$

or $T = 28°C$

Energy distribution of Black Body Radiation

We draw a graph showing the energy distribution with wavelength at different temperatures.



It is seen that the energy emitted corresponding to a particular wavelength is maximum. The area enclosed by the above graph with x-axis given the total radiant power, which is proportional to the fourth power of the temperature.

Wien's Displacement Law

The wavelength (λ_m) of maximum intensity of emission of black body radiation is inversely proportional to absolute temperature (T) of the black body.

$$\lambda_m \propto \frac{1}{T}$$

$$\lambda_m = \frac{b}{T}$$

$$b = 2.898 \times 10^{-3}\ mK$$